# MOTIVIC HOMOTOPY THEORY

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Abstract. These unofficial notes were prepared by Yiming Wang based on the lecture course "Motivic homotopy theory," delivered by Marc Hoyois at the University of Regensburg during the winter semester 2026–2027. Be cautious that these notes need revision and are rough in many places. All mistakes and misconceptions are due to the author.

*AI Disclosure*: The author used ChatGPT 6 Astra to refine the language and review the mathematical content, particularly in the introductory section on synthetic category theory.

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## 1. Introduction

Throughout this section, let $k$ be a field and let $\mathrm{SmProp}_k$ be the category of smooth proper $k$-schemes.

**Grothendieck's motives.** Grothendieck proposed the concept of *motives* in the 1960s to explain the common features of the *Weil cohomology* theories of algebraic varieties.

*Example* 1.1 (Weil cohomology theories). Let $X \in \mathrm{SmProp}_k$.

(a) (Betti cohomology) Let $\iota\colon k \hookrightarrow \mathbf{C}$ be an embedding. Consider the *singular cohomology* of $X(\mathbf{C})$ with the analytic topology $\mathrm{H}^*_{\mathrm{Betti}}(X) = \mathrm{H}^*(X(\mathbf{C}), \mathbf{Q})$. This defines a functor
$$\mathrm{H}^*_{\mathrm{Betti}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_{\mathbf{Q}}^{\mathrm{gr}}$$
depending on the embedding.

(b) (Algebraic de Rham cohomology) Suppose that $\mathrm{char}(k) = 0$. The *algebraic de Rham complex* defines cohomology groups $\mathrm{H}^*_{\mathrm{dR}}(X/k)$ equipped with the Hodge filtration. There are functors
$$\mathrm{H}^*_{\mathrm{dR}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_k^{\mathrm{gr}},$$

(c) (Étale cohomology) Let $k^{\mathrm{sep}}$ be a separable closure of $k$, $G_k = \mathrm{Gal}(k^{\mathrm{sep}}/k)$, and let $\ell \neq \mathrm{char}(k)$ be a prime. The *étale cohomology* groups $\mathrm{H}^*_{\text{ét}}(X_{k^s}, \mathbf{Q}_\ell)$ give us the functors
$$\mathrm{H}^*_{\text{ét}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_{\mathbf{Q}_\ell}^{\mathrm{gr}}.$$
This construction does not require an embedding into $\mathbf{C}$ and applies in positive characteristic.

(d) (Crystalline cohomology) Let $k$ be a perfect field of characteristic $p > 0$. The *crystalline cohomology* groups $\mathrm{H}^*_{\mathrm{cris}}(X/\mathrm{W}(k))[\frac{1}{p}]$ give us the functors
$$\mathrm{H}^*_{\mathrm{cris}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_{\mathrm{W}(k)[\frac{1}{p}]}^{\mathrm{gr}},$$

Certain features suggest that these cohomology theories can be compared, at least to some extent.

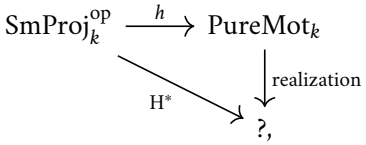
*Remark* 1.2 (Comparison between Weil cohomology theories). The Weil cohomology theories in Example 1.1 are related by comparison isomorphisms. Let $k \hookrightarrow \mathbf{C}$ and $k^{\mathrm{sep}} \hookrightarrow \mathbf{C}$ be embeddings. There are natural isomorphisms
$$\mathrm{H}^*_{\mathrm{Betti}}(X) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \simeq \mathrm{H}^*_{\text{ét}}(X_{k^{\mathrm{sep}}}, \mathbf{Q}_\ell) \quad \text{and} \quad \mathrm{H}^*_{\mathrm{Betti}}(X) \otimes_{\mathbf{Q}} \mathbf{C} \simeq \mathrm{H}^*_{\mathrm{dR}}(X/k) \otimes_k \mathbf{C}.$$
The difficulty lies in the fact that the theories do not take values in vector spaces over a common field.

Further evidence is provided by numerical invariants. For example,

**Theorem 1.3** (Independence of $\ell$)**.** *Let $X$ be a smooth and proper scheme over $k$. The $i$-th* Betti number
$$b_i(X) = \dim_{\mathbf{Q}_\ell} \mathrm{H}^i_{\text{ét}}(X_{k^s}, \mathbf{Q}_\ell)$$
*is independent of the prime $\ell \neq \mathrm{char}(k)$.*

*Remark* 1.4. It remains an open question whether Theorem 1.3 extends to arbitrary schemes of finite type over a field. This is one of the celebrated *independence-of-$\ell$ conjectures*.

Grothendieck's idea was to account for these similarities by a category of *pure motives* over $\mathbf{Q}$, together with realization functors recovering the different cohomology theories. The proposed picture is




where ? is understood as the cohomological coefficients and $h(X)$ is a universal linear object attached to $X$.

*Tannakian duality* asserts that every symmetric monoidal category satisfying the Tannakian axioms is equivalent to the category of representations of an affine *gerbe*. Conjecturally, a suitable category of pure motives, with an appropriately modified commutativity constraint, is Tannakian over $\mathbf{Q}$. Tannakian duality then describes it in terms of a motivic Galois gerbe. The cohomological invariants discussed above are understood as different realizations of underlying objects in the category of motives.[1]

Although the theory of pure motives was originally developed in part to address the Weil conjectures, it remains a fruitful idea in algebraic geometry, even after the conjectures were proved by other methods.

**Stable homotopy theory.** We now pass from graded cohomology groups to complexes and spectra. This suggests a homotopical refinement of the motivic viewpoint. A "cohomology theory" on a "geometric category" $\mathcal{G}$ is a contravariant functor

$$E\colon \mathcal{G}^{\mathrm{op}} \longrightarrow \mathcal{C},$$

where $\mathcal{C}$ is a stable category. Examples of $\mathcal{G}$ include topological spaces, manifolds, and schemes; examples of $\mathcal{C}$ include spectra Sp, module spectra, filtered spectra, and the derived categories $\mathrm{D}(\mathbf{Z})$ and $\mathrm{D}(k)$.

The following are some examples of cohomology theories:

*Example* 1.5 (Cohomology theories)*.*

(a) (Ordinary cohomology) Singular cochains define a functor

$$\mathrm{C}^*(-,\mathbf{Z})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{D}(\mathbf{Z}).$$

For a locally contractible paracompact Hausdorff space $X$, this agrees with $\mathrm{R}\Gamma(X,\mathbf{Z}_X)$, the derived global sections of the constant sheaf.

(b) (Complex topological $K$-theory) The spectrum KU defines a functor

$$\Gamma(-,\mathrm{KU})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{Sp}, \quad \Gamma(X,\mathrm{KU}) = \mathrm{map}_{\mathrm{Sp}}(\Sigma^\infty_+ X, \mathrm{KU}),$$

its homotopy group gives us the *complex topological $K$-theory* groups

(c) (Complex cobordism) The *complex cobordism spectrum* MU defines

$$\Gamma(-,\mathrm{MU})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{Sp}, \quad \Gamma(X,\mathrm{MU}) = \mathrm{map}_{\mathrm{Sp}}(\Sigma^\infty_+ X, \mathrm{MU}).$$

its homotopy groups gives us the complex cobordism groups.

(d) (Étale cohomology) Let $\ell$ be a prime. On schemes on which $\ell$ is invertible, we have

$$\mathrm{R}\Gamma_{\text{ét}}(-,\mathbf{Z}/\ell)\colon \mathrm{Sch}^{\mathrm{op}}_{\mathbf{Z}[\frac{1}{\ell}]} \to \mathrm{D}(\mathbf{Z}/\ell).$$

Let $X$ be a scheme of finite type over $k$, where $\mathrm{char}(k) \neq \ell$. Its $\ell$-adic cohomology complexes are

$$\mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}_\ell) = \lim_{n\geq 1} \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}/\ell^n), \quad \text{and} \quad \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Q}_\ell) = \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}_\ell) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell,$$

(e) (Algebraic $K$-theory) Let $\mathrm{Sch}_{\mathrm{qcqs}}$ be the category of quasi-compact quasi-separated schemes. *Nonconnective algebraic $K$-theory* gives a functor

$$\mathrm{K}\colon \mathrm{Sch}^{\mathrm{op}}_{\mathrm{qcqs}} \to \mathrm{Sp}, \quad X \mapsto \mathrm{K}(\mathrm{Perf}(X)),$$

Let $R$ be a commutative ring and $\mathrm{Vect}_R$ be the category of finitely generated projective $R$-modules. Then

$$\Omega^\infty \mathrm{K}(\mathrm{Spec}(R)) \simeq (\mathrm{Vect}_R^{\simeq})^{\mathrm{grp}},$$

where $(-)^{\mathrm{grp}}$ is the group completion functor.

In fact, cohomology theories in the examples above carries more structure than just functoriality.

*Example* 1.6 (Additional structures)*.* Let $X$ be a topological space.

[1]Although there are some possible categories for pure motives, their expected Tannakian properties remain conjectural in general. Grothendieck's *standard conjectures* on algebraic cycles would supply the necessary compatibility with cohomological realizations.

(a) (Cup product) The cup product is a map
$$- \smile -\colon \mathrm{H}^i(X, \mathbf{Z}) \otimes \mathrm{H}^j(X, \mathbf{Z}) \to \mathrm{H}^{j+i}(X, \mathbf{Z}),$$
which makes $\mathrm{H}^*(X, \mathbf{Z})$ into a graded-commutative ring.

(b) (Twisted coefficients) Let $A$ be a local system of abelian groups on $X$. There is the notion of cohomology $\mathrm{H}^*(X, A)$ and homology $\mathrm{H}_*(X, A)$ with coefficients in $A$.

(c) (Wrong-way functoriality) Let $f\colon Y \to X$ be a proper smooth submersion of manifolds without boundary, of relative dimension $d$, and let $\mathrm{or}_f$ denote its relative orientation local system. Integration along the fibers defines a *Gysin pushforward*
$$f_!\colon \mathrm{H}^{i+d}(Y, \mathrm{or}_f) \longrightarrow \mathrm{H}^i(X, \mathbf{Z}).$$
A choice of relative orientation identifies $\mathrm{or}_f \simeq \mathbf{Z}_Y$, yielding an untwisted pushforward.

*Example* 1.7 (Poincaré duality)*.* Let $X$ be an $n$-manifold without boundary, with orientation local system $\mathrm{or}_X$. Poincaré duality gives
$$\mathrm{H}^i(X, \mathbf{Z}) \simeq \mathrm{H}^{\mathrm{BM}}_{n-i}(X, \mathrm{or}_X), \qquad \mathrm{H}^i_c(X, \mathbf{Z}) \simeq \mathrm{H}_{n-i}(X, \mathrm{or}_X),$$
where $\mathrm{H}^{\mathrm{BM}}_*$ denotes *Borel–Moore homology*. The dualizing complex is $\omega_X \simeq \mathrm{or}_X[n]$.

*Example* 1.8 ($G$-theory)*.* Let $X$ be a regular noetherian scheme of finite Krull dimension. The canonical map
$$\mathrm{K}(X) \simeq \mathrm{G}(X).$$
is an isomorphism. This is a $K$-theoretical analogue of Poincaré duality.

**Six functor formalism.** Grothendieck's fundamental insight was to organize cohomology theories categorically, replacing cohomology groups by categories of coefficients that are *bivariantly functorial*.

The category of coefficients for ordinary cohomology on a topological space $X$ is the category of sheaves $\mathrm{Shv}(X, \mathrm{D}(\mathbf{Z}))$. For every continuous map $f\colon Y \to X$, there is a *pullback* functor
$$f^*\colon \mathrm{Shv}(X, \mathrm{D}(\mathbf{Z})) \to \mathrm{Shv}(Y, \mathrm{D}(\mathbf{Z})).$$
where $U \subseteq X$ is an open subset. It admits a right adjoint $f_*$, i.e., the *pushforward* functor, it is explicitly given by $(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U))$.

Suppose that $Y$ and $X$ are locally compact Hausdorff spaces. There is the *pushforward of proper support* $f_!\colon \mathrm{Shv}(Y, \mathrm{D}(\mathbf{Z})) \to \mathrm{Shv}(X, \mathrm{D}(\mathbf{Z}))$. The formula for $f_!$ is slightly more complicated:
$$(f_!\mathcal{F})(U) = \operatorname*{colim}_{K \subseteq f^{-1}(U)} \mathrm{fib}\left(\mathcal{F}\left(f^{-1}(U)\right) \to \mathcal{F}\left(f^{-1}(U) \setminus K\right)\right),$$
where the colimit ranges over closed subsets $K \subseteq f^{-1}(U)$ such that $f|_K\colon K \to U$ is proper. The functor $f_!$ admits a right adjoint $f^!$, the *exceptional pullback functor*

The above datum are related by the *base change isomorphism*.

**Theorem 1.9** (Proper base change)**.** *Consider a cartesian square of locally compact Hausdorff spaces*
$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X. \end{array}$$
*There exists a canonical isomorphism $g^* f_! \simeq f'_! h^*$.*

*Remark* 1.10*.* Let $X$ be a locally compact Hausdorff space, and let $p\colon X \to *$.

(a) The *cohomology* of $X$ with coefficients in $\mathbf{Z}$ is given by the pushforward $\Gamma(X, \mathbf{Z}) = p_*\mathbf{Z}$.

(b) The *cohomology with compact support* of $X$ with coefficients in $\mathbf{Z}$ is given by the pushforward with proper support $\Gamma_c(X, \mathbf{Z}) = p_!\mathbf{Z}$.

(c) Writing $\omega_X = p^!\mathbf{Z}$ for the dualizing sheaf, the *homology* of $X$ with coefficients in $\mathbf{Z}$ is given by $p_!\omega_X = p_!p^!\mathbf{Z}$.

The above data can be incorperated into the category of *spans*.

**Definition 1.11** (Category of spans)**.** Let $\mathcal{G}$ be a category with finite limits. Its *category of spans* $\mathrm{Span}(\mathcal{G})$ is defined as follows:

(a) the objects are those of $\mathcal{G}$;
(b) a morphism $X \to Y$ is a span $X \xleftarrow{f} Z \xrightarrow{g} Y$.
(c) the composite of $X \leftarrow Z \to Y$ and $Y \leftarrow T \to W$ is represented by $X \leftarrow Z \times_Y T \to W$.
(d) the identity of $X$ is represented by $X \xleftarrow{\mathrm{id}} X \xrightarrow{\mathrm{id}} X$.

Let LCH be the category of locally compact Hausdorff spaces. The sheaf operations give a functor

$$\mathrm{Shv}(-, \mathrm{D}(\mathbf{Z})) \colon \mathrm{Span}(\mathrm{LCH}) \longrightarrow \widehat{\mathrm{Cat}}, \quad \left(X \xleftarrow{f} Z \xrightarrow{g} Y\right) \mapsto g_!f^*.$$

Its compatibility with composition is exactly the compatibility expressed by base change.

**Definition 1.12** (Presentable six-functor formalism)**.** Let $\mathcal{G}$ be a category with finite limits. A *presentable six-functor formalism* on $\mathcal{G}$ is a lax symmetric monoidal functor

$$\mathrm{Span}(\mathcal{G}) \longrightarrow \mathrm{Pr}^L,$$

where $\mathrm{Pr}^L$ is the *category of presentable categories* and colimit-preserving functors, with the *Lurie tensor product*.

The topological example gives a presentable six-functor formalism, i.e., there exists a functor

$$\mathrm{Shv}(-, \mathrm{Sp}) \colon \mathrm{Span}(\mathrm{LCH}) \longrightarrow \mathrm{Pr}^L.$$

In algebraic geometry, two "universal" sources of six-functor formalisms are *motivic homotopy theory*, developed by Morel and Voevodsky, and the theory of *analytic stacks*, developed by Clausen and Scholze. The former encompasses many of the cohomology theories discussed above, including étale cohomology and variants of $K$-theory, whereas the latter provides a six-functor formalism for quasi-coherent sheaves.[2]

**Motivic homotopy theory.** Morel and Voevodsky's construction gives a functor

$$\mathrm{SH}(-) \colon \mathrm{Sch}^{\mathrm{op}} \longrightarrow \mathrm{Pr}^L, \qquad X \longmapsto \mathrm{SH}(X),$$

where $\mathrm{SH}(X)$ is the *stable* $\mathbf{A}^1$*-homotopy category* over $X$. Its objects are called *motivic spectra*. This is the algebraic analogue of the six functor formalism

$$\mathrm{Shv}(-, \mathrm{Sp}) \colon \mathrm{Top}^{\mathrm{op}} \longrightarrow \mathrm{Pr}^L, \quad X \longmapsto \mathrm{Shv}(X, \mathrm{Sp}).$$

**Theorem 1.13** (Ayoub)**.** *On qcqs schemes,* SH *admits a six-functor formalism, i.e. a lax symmetric monoidal functor*

$$\mathrm{Span}_{\mathrm{sep,ft}}(\mathrm{Sch}_{\mathrm{qcqs}}) \longrightarrow \mathrm{Pr}^L, \qquad X \longmapsto \mathrm{SH}(X).$$

*where the source is the category of spans whose right leg is separated and of finite type.*

One goal of this course is to understand this formalism. Morel and Voevodsky's construction rests on three theorems: *homotopy purity* and *localization*, due to Morel and Voevodsky, and *ambidexterity*, due to Ayoub and Röndigs. The first two are *unstable* results.

In topology, one studies the relative manifolds over a topological space. The scheme-theoretical analogue of relative manifolds are *smooth schemes*.

**Definition 1.14** (Relative manifolds)**.** Let $X$ be a topological space. A *relative manifold* over $X$ is a map $Y \to X$ of topological spaces locally isomorphic to a projection $D^n \times U \to U$, where $U \subset X$ is an open subset.

[2] A natural question is whether these two approaches can be unified. Moreover, a related question is to what extent algebraic $K$-theory can be incorporated into the six-functor formalism of motivic homotopy theory. At present, the available formalism for algebraic $K$-theory is only partial, although a variant of the theory fits into this framework.

Stable homotopy theory provides a setting for cohomology theories. Unstable homotopy theory also retains nonabelian information, such as the fundamental group. Historically, this viewpoint was motivated in part by cohomology with nonabelian coefficients. In Definition 1.12, the category $\mathrm{Pr}^L$ accommodates such nonabelian information.

Passing to homotopy-invariant sheaves on relative manifolds over $X$ recovers $\mathrm{Shv}(X, \mathrm{An})$. Analogously, in algebraic geometry, the category $\mathrm{H}(X)$ of $\mathbf{A}^1$*-invariant Nisnevich sheaves* on smooth $X$-schemes provides the unstable counterpart of $\mathrm{SH}(X)$. To pass from $\mathrm{H}(X)$ to $\mathrm{SH}(X)$, one needs to add the Tate twist. Therefore, $\mathrm{SH}(X)$ is not the naive categorical stabilization of $\mathrm{H}(X)$.

The analogy can be summarized as follows:

| Topological spaces | Schemes |
|---|---|
| $\mathrm{Shv}(X, \mathrm{Sp})$ | $\mathrm{SH}(X)$ |
| $\mathrm{Shv}(X, \mathrm{An})$ | $\mathrm{H}(X)$ |
| Relative manifolds over $X$ | Smooth schemes over $X$ |
| Ordinary cohomology H$\mathbf{Z}$ | Motivic cohomology H$\mathbf{Z}$ |
| Complex $K$-theory KU | Homotopy $K$-theory KGL |
| Complex cobordism MU | Algebraic cobordism MGL |

**Synthetic category theory.** In these notes, we will freely speak of (higher) category theory. Here we will explain the basic ideas of homotopy theory and its relation to the foundations of mathematics.

Set theory provides a logical foundation, but its language does not directly reflect several features of mathematical practice.

First, every object of set theory is a set, and equality can be asked of any two objects. In classical logic, they are either equal or unequal. Mathematical language, by contrast, is usually *strongly typed*: one speaks of an integer, a group, or a manifold. *Type theory* makes this explicit by considering each term with a specified type. Unlike set theory, where any two elements can be collected into a set, type theory does not automatically place terms of different types in a common type.

Type conversions are nevertheless ubiquitous. A map between types sends a term of the source to a term of the target. For example, the integers and the even integers both appear to contain 2. In type theory, the inclusion of even integers into integers identifies the images of these terms in a common type. In this sense, the two types have an element in common.

Second, the usual formulation of set theory uses first-order logic, whose variables range over all sets. Mathematical statements generally use restricted quantifiers, such as "for every manifold $X$." Such restrictions can be expressed in set theory, whereas type theory incorporates them into the language.

Third, many collections of mathematical objects are not naturally set-like. If $A$ is a set and $x, y \in A$, the equality $x = y$ is a truth value: classically, it is either true or false. If $x$ and $y$ are objects of a category, however, there may be several isomorphisms between them. Each gives a way of identifying the objects and transporting their categorical properties. If these identifications are regarded as equalities, equality carries more information than a truth value.

The homotopical viewpoint is to regard identifications themselves as mathematical objects. For $x, y$ in such a collection $A$, there is a collection $\mathrm{Eq}(x, y)$ of identifications, which in turn has its own identifications. This is a fundamental principle of homotopy theory. An object with this structure is called an *anima*, a *space*, or an $\infty$*-groupoid*; in type theory, it is interpreted as a *type*. The collection of animae form a category An.

A set is a special case: for each $x, y \in A$, the anima $\mathrm{Eq}(x, y)$ is empty or contractible, expressing that equality is false or true. Here contractibility means uniqueness up to all higher identifications. An anima $A$ is a *truth value* if $\mathrm{Eq}(x, y)$ is contractible for every $x, y \in A$. These distinctions are the first instances of truncation. We will explain more about these ideas in Section 2.

**Notation 1.15.** In these notes, we adopt the following notation:

(a) (ordinary) rings refer to commutative rings in the classical sense, which differs from animated (commutative) rings.

## 2. Preliminaries

This section presents a naive and informal approach to algebraic geometry and (higher) category theory. It is intended solely to provide intuition and should not be regarded as a rigorous or systematic treatment of the subject.[3]

### Category theory.

**Definition 2.1.** A functor $f\colon \mathcal{C} \to \mathcal{D}$ is called

(a) *fully faithful* if the canonical map

$$\mathrm{Map}_{\mathcal{C}}(x, y) \longrightarrow \mathrm{Map}_{\mathcal{D}}(f(x), f(y))$$

is an equivalence for all $x, y \in \mathcal{C}$;

(b) a *localization* if, for every category $\mathcal{E}$, precomposition with $f$ identifies $\mathrm{Fun}(\mathcal{D}, \mathcal{E})$ with the full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathcal{E})$ consisting of functors sending $W$ to equivalences. We then write $\mathcal{D} \simeq \mathcal{C}[W^{-1}]$.

**Proposition 2.2.** *Let $f\colon \mathcal{D} \to \mathcal{C}$ be a functor with a left or right adjoint $g\colon \mathcal{C} \to \mathcal{D}$. The following are equivalent:*

*(a) $f$ is fully faithful;*

*(b) $g$ is a localization.*

*Proof.* Exercise. □

**Definition 2.3** (Bousfield localization)**.** In Proposition 2.2, assume that $f$ is fully faithful. We call $\mathcal{D}$ a *left Bousfield localization* of $\mathcal{C}$ if $g$ is left adjoint to $f$, and a *right Bousfield localization* if $g$ is right adjoint to $f$.

*Convention* 2.4. Consider the following adjunction diagram, i.e., $f$ is the left adjoint of $g$:

$$\mathcal{C} \underset{g}{\overset{f}{\rightleftarrows}} \mathcal{D}.$$

We will always write the left adjoint on top of the right adjoint.

*Example* 2.5 (Examples of Bousfield localizations)*.*

(a) The inclusion $\mathrm{Ab} \hookrightarrow \mathrm{Grp}$ of abelian groups into groups has a left adjoint, called *abelianization*. Thus, Ab is a left Bousfield localization of Grp.

(b) The inclusion $\mathrm{Ab} \hookrightarrow \mathrm{CMon}$ has both a left and a right adjoint. Thus, Ab is both a left and a right Bousfield localization of CMon.

(c) The inclusion $\mathrm{Mod}_{\mathbb{Q}} \hookrightarrow \mathrm{Ab}$ has both a left and a right adjoint. Thus, $\mathrm{Mod}_{\mathbb{Q}}$ is both a left and a right Bousfield localization of Ab.

*Exercise* 2.6. Show that a fully faithful functor is a monomorphism in Cat and that a localization is an epimorphism. Do the converses hold?

**Definition 2.7** (Truncated animae)**.** Let $A$ be an anima.

(a) $A$ is $(-2)$*-truncated* if $A \simeq *$, i.e., if $A$ is contractible;

(b) $A$ is $n$*-truncated*,[4] for an integer $n \geq -1$, if $\mathrm{Eq}(x, y)$ is $(n-1)$-truncated for all $x, y \in A$.

*Remark* 2.8 (Truncation)*.*

(a) For $-2 \leq n \leq \infty$, let $\mathrm{An}_{\leq n} \subseteq \mathrm{An}$ be the full subcategory of $n$-truncated anima. Its inclusion has a left adjoint $\tau_{\leq n}$, called $n$*-truncation*;

(b) the $(-1)$-truncation functor $\tau_{\leq -1}$ is left adjoint to the inclusion of truth values;

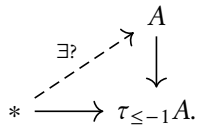
(c) the 0-th truncation functor $\tau_{\leq 0}$ sends an anima to its set of connected components.

[3] These are the comments of the note-taker, i.e. Yiming Wang.

[4] An $n$-truncated anima is also called an $n$*-groupoid*.

*Remark* 2.9 (Type theoretical interpretation). In type theory, *propositions* are $(-1)$-truncated anima. To prove a proposition is to construct a term of that type. For an anima $A$, the truth value $\tau_{\leq -1}A$ expresses that $A$ is nonempty. An *term* or an *element* of $\tau_{\leq -1}A$ is a proof of this proposition.

Constructing a term of $A$ gives such a proof, but a proof of mere existence does not in general specify a term of $A$. Type theory thus makes explicit the distinction between proving existence and exhibiting a witness. More precisely, recovering a witness from a proof of existence is a lifting problem:




The lifts themselves form an anima.

*Remark* 2.10. Many animae are not be $n$-truncated for any $n \geq -1$, even when obtained by simple constructions. For example, define the *$n$-sphere* $S^n$ inductively by

$$S^{-1} = \varnothing \quad \text{and} \quad S^n = \Sigma S^{n-1}$$

where $n \geq 0$ and $\Sigma X = * \sqcup_X *$ is the *suspension* of $X$, where the pushout is taken in An.[5]

In particular, $S^0$ consists of two points, and $S^1$ is obtained by adjoining two identifications between two points. For every $n \geq 2$, the sphere $S^n$ is not $m$-truncated for any finite $m \geq -1$.[6]

*Exercise* 2.11. The map $\eta\colon S^3 \to S^2$ is called the *Hopf fibration*. How can one interpret the equivalence

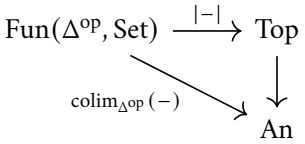
$$\tau_{\leq 0} \operatorname{Map}(S^3, S^2) \simeq \mathbf{Z}\eta,$$

where $\eta$ is the *Hopf element*, from the perspective of type theory? What does $\eta$ represent in the corresponding type, and why does it exist?

The functor Top $\to$ An is a localization at weak homotopy equivalences. Classical invariants developed in the twentieth century, such as singular cohomology and homotopy groups, factor through this localization and thus study anima indirectly. There is also a localization from *simplicial sets*, again at weak homotopy equivalences

$$\operatorname{Fun}(\Delta^{\mathrm{op}}, \mathrm{Set}) \to \mathrm{An}, \quad K \mapsto \operatorname*{colim}_{\Delta^{\mathrm{op}}} K,$$

called *geometric realization*.[7] Moreover, the following diagram commutes




where $|-|:\ \operatorname{Fun}(\Delta^{\mathrm{op}}, \mathrm{Set}) \to \mathrm{Top}$ is the topological geometric realization.

**Definition 2.12.** The notion $(n,k)$-*categories* is defined inductively as follows:

(a) an $(\infty, 0)$-*category* is an anima;
(b) an $(n, 0)$-*category*, for $-2 \leq n < \infty$, is an $n$-truncated anima;
(c) an $(n,k)$-*category*, where $k \geq 1$, is a category enriched in $(n-1, k-1)$-categories;

This gives the range $-2 \leq n \leq \infty$ and $0 \leq k \leq n+2$, with $k$ finite. There are also theories of $(\infty,\infty)$-categories obtained as suitable limits of the theories of $(\infty, n)$-categories.

*Example* 2.13.

[5]Intuitively, $\Sigma X$ is obtained from two points by adjoining identifications between them parametrized by $X$.
[6]It remains unknown whether this fact can be proved purely type-theoretically.
[7]The fact that it is a localization is nontrivial.

(a) $(0,0)$-categories are sets;
(b) $(0,1)$-categories are posets;
(c) $(1,2)$-categories are categories enriched in posets;
(d) $(1,1)$-categories are ordinary categories.

*Exercise* 2.14. For $n \geq -1$, what is an $(n, n+2)$-category?

Traditionally, an $n$-category means an $(n,n)$-category; in modern terms, it often means an $(\infty, n)$-category. The term $\infty$-category classically refers to a theory of $(\infty,\infty)$-categories; in modern terms, it also often means an $(\infty,1)$-category. In these notes, a *category* means an $(\infty,1)$-category and *ordinary categories* refer to $(1,1)$-categories.[8]

As for ordinary categories, we have objects, morphisms, identities, and composition, but their collections are now animae. For a category $\mathcal{C}$, write $\mathcal{C}_0$ for its anima of *objects* and $\mathcal{C}_1$ for its anima of *morphisms*. There are source and target maps $s,t\colon \mathcal{C}_1 \rightrightarrows \mathcal{C}_0$, an identity map $x \mapsto \mathrm{id}_x$, and a composition map

$$\mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1 \longrightarrow \mathcal{C}_1,$$

where $\mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1$ is the pullback of $s$ along $t$. Furthermore, we require associativity and the unitality, i.e., there exists identifications such that

$$(f \circ g) \circ h \simeq f \circ (g \circ h) \quad \text{and} \quad f \circ \mathrm{id} \simeq f \simeq \mathrm{id} \circ f.$$

For four composable morphisms, the two chains of equalities from $((fg)h)k$ to $f(g(hk))$ must themselves be identified via a 2-*equality*. Longer composites require higher equalities. These compatibility data are the *higher coherences* of composition of morphisms.

There are several ways to encode higher coherences. For example, Stasheff's *associahedra* explicitly organize them. A more convenient encoding uses the *unbiased* viewpoint. For every $n \geq 0$, there exists a collection of $n$-ary operations. For example, a 3-ary composite $f \circ g \circ h$ is compared with both $(f \circ g) \circ h$ and $f \circ (g \circ h)$.

These data can be organized by the following category. Its objects are finite linearly ordered sets, including the empty set, and a morphism $I \longrightarrow J$ is an order-preserving map

$$I \longrightarrow \{-\infty\} \sqcup J \sqcup \{+\infty\}.$$

Composition is defined by extending maps to fix $\pm\infty$. In this model, write $[n] = \{1 < \cdots < n\}$, with $[0] = \varnothing$.

*Exercise* 2.15. Show that this category is equivalent to the opposite of the simplex category $\Delta^{\mathrm{op}}$.

Under this identification, the constant map $[n] \to [1]$ with value 1 encodes the $n$-ary composition. The map $[3] \to [1]$ has two factorizations through an order-preserving surjection $[3] \twoheadrightarrow [2]$, corresponding to the two parenthesizations of a ternary composite.

For example, the associative multiplication of a group can be encoded simplicially. More precisely, there exists a fully faithful functor $\mathrm{Grp} \hookrightarrow \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Set})$.

A category is similarly encoded by a simplicial anima $\mathcal{C}\colon \Delta^{\mathrm{op}} \to \mathrm{An}$. Its values $\mathcal{C}_0$, $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_3$ are the anima of objects, morphisms, composable pairs, and composable triples. Moreover,

$$\mathcal{C}_2 \simeq \mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1, \qquad \mathcal{C}_3 \simeq \mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1.$$

Not every simplicial anima has these properties. In fact, categories are *Segal animae*:

**Definition 2.16** (Segal object)**.** Let $\mathcal{E}$ be a category with finite limits. A simplicial object $\mathcal{C} \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{E})$ is a *Segal object*, or *precategory*, if, for every $n \geq 2$, the map

$$\mathcal{C}_n \longrightarrow \underbrace{\mathcal{C}_1 \times_{\mathcal{C}_0} \cdots \times_{\mathcal{C}_0} \mathcal{C}_1}_{n \text{ factors}}$$

[8] Since the lectures were held in Regensburg…

is an equivalence. Its components are induced by the maps $\rho_i\colon [n] \to [1]$ sending $j < i$ to $-\infty$, $j = i$ to 1, and $j > i$ to $+\infty$. Binary composition is the map

$$\mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1 \simeq \mathcal{C}_2 \longrightarrow \mathcal{C}_1$$

induced by the constant map $[2] \to [1]$ with value 1.

*Example* 2.17. A 20-*th century category* in the classical, set-based sense is exactly a Segal set. [9]

A category is more than a Segal anima: identifications of objects must agree with equivalences. This condition is called *univalence*. One could achieve univalence by localizing the category of Segal anima at the equivalences, but this does not give explicit descriptions of the objects in the localization.

Indeed, for a Segal anima $\mathcal{C}$, let $\mathrm{Iso}(\mathcal{C}) \subseteq \mathcal{C}_1$ be the subanima of invertible morphisms. The identity map $\mathcal{C}_0 \to \mathcal{C}_1$ factors through $\mathrm{Iso}(\mathcal{C})$.

**Definition 2.18** (Univalence)**.** A Segal anima $\mathcal{C}$ is *univalent*, or *Rezk-complete*, if the identity map

$$\mathcal{C}_0 \longrightarrow \mathrm{Iso}(\mathcal{C}), \quad x \mapsto \mathrm{id}_x$$

is an equivalence.

**Definition 2.19** (Category)**.** A *category* is a univalent Segal anima. We denote by Cat the category of categories.

**Definition 2.20.** Let $f\colon \mathcal{C} \to \mathcal{D}$ be a map of Segal anima.

(a) $f$ is *fully faithful* if, for all $x, y \in \mathcal{C}_0$, the map

$$\mathrm{Map}_{\mathcal{C}}(x, y) \longrightarrow \mathrm{Map}_{\mathcal{D}}(f(x), f(y))$$

is an equivalence. [10].

(b) $f$ is *essentially surjective* if, for every $d \in \mathcal{D}_0$, there are $c \in \mathcal{C}_0$ and an isomorphism $f(c) \simeq d$.

*Remark* 2.21. There are left Bousfield localizations

$$\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{An}) \underset{\hookrightarrow}{\overset{\mathrm{L_{Seg}}}{\leftarrow}} \mathrm{Fun}_{\mathrm{Seg}}(\Delta^{\mathrm{op}}, \mathrm{An}) \underset{\hookrightarrow}{\overset{\mathrm{L_{univ}}}{\leftarrow}} \mathrm{Cat},$$

where the functors $\mathrm{L_{Seg}}$ and $\mathrm{L_{univ}}$ universally impose the Segal and univalence conditions respectively. Moreover, $\mathrm{L_{univ}}$ is localization at the fully faithful essentially surjective maps.

*Example* 2.22 (Segal sets)*.* The inclusion $\mathrm{Set} \hookrightarrow \mathrm{An}$ gives us the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Fun}_{\mathrm{Seg}}(\Delta^{\mathrm{op}}, \mathrm{Set}) & \hookrightarrow & \mathrm{Fun}_{\mathrm{Seg}}(\Delta^{\mathrm{op}}, \mathrm{An}) \\ \downarrow{\scriptstyle \mathrm{L_{Seg}}} & & \downarrow{\scriptstyle \mathrm{L_{Seg}}} \\ \mathrm{Cat}_{(1,1)} & \hookrightarrow & \mathrm{Cat}. \end{array}$$

where the left vertical map is given by the localization at the fully faithful essentially surjective maps. Nevertheless, the right vertical map is not a Bousfield localization when restricted to Segal sets, hence does not land in $\mathrm{Cat}_{(1,1)}$.

*Exercise* 2.23.

(a) What are univalent Segal $(1,1)$-categories?

(b) Explain the relation between univalence and fully faithful essentially surjective functors. More precisely, show that a Segal anima $\mathcal{C}$ is univalent if and only if, for every fully faithful essentially surjective map of Segal animae $f\colon \mathcal{D} \to \mathcal{E}$, the restriction map

$$f^*\colon \mathrm{Map}(\mathcal{E}, \mathcal{C}) \longrightarrow \mathrm{Map}(\mathcal{D}, \mathcal{C})$$

is an equivalence.

[9] A 20-th century category is *not* the same as an ordinary category.

[10] Here $\mathrm{Map}_{\mathcal{C}}(x, y)$ is the fiber of $(s, t)\colon \mathcal{C}_1 \to \mathcal{C}_0 \times \mathcal{C}_0$ at $(x, y)$, i.e., $\{x\} \times_{\mathcal{C}_0} \mathcal{C}_1 \times_{\mathcal{C}_0} \{y\}$

*Remark* 2.24 (Homotopy $n$-category). For $n \geq 1$, the inclusion $\mathrm{Cat}_{(n,1)} \hookrightarrow \mathrm{Cat}$ has a left adjoint $\mathrm{h}_n$, the *homotopy $n$-category*. For $x, y \in \mathcal{C}_0$, it satisfies $(\mathrm{h}_n\mathcal{C})_0 \simeq \tau_{\leq n}(\mathcal{C}_0)$ and

$$\mathrm{Map}_{\mathrm{h}_n\mathcal{C}}(\bar{x}, \bar{y}) \simeq \tau_{\leq n-1} \mathrm{Map}_{\mathcal{C}}(x, y),$$

where $\bar{x}$ and $\bar{y}$ denote the images of $x$ and $y$ under $\mathcal{C}_0 \to \tau_{\leq n}(\mathcal{C}_0)$.

*Remark* 2.25 (Animae as categories). The constant simplicial anima functor gives an inclusion

$$\mathrm{cst}\colon \mathrm{An} \hookrightarrow \mathrm{Cat} \subseteq \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{An}).$$

Its left adjoint is the *geometric realization* functor $|-|$, given by $|\mathcal{C}| = \mathrm{colim}_{[n]\in\Delta^{\mathrm{op}}} \mathcal{C}_n$, and its right adjoint is the *core* functor $(-)^{\simeq}$, given by $\mathcal{C}^{\simeq} \simeq \lim_{\Delta^{\mathrm{op}}} \mathcal{C}$. The core is the maximal anima containing all objects and only the invertible morphisms. The unit map $\mathcal{C} \to \mathrm{cst}(|\mathcal{C}|)$ is localization at all morphisms.

We now discuss constructions of categories. Among the categories already encountered are Cat, $\mathrm{Cat}_{(1,1)}$, and Set. By Remark 2.21, the category Cat admits small limits and colimits. Its inclusion into $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{An})$ preserves limits and filtered colimits, but not colimits in general.

**Construction 2.26.** Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

(a) There is a *functor category* $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$.
(b) For $X \in \mathcal{C}$, there are *slice categories* $\mathcal{C}_{/X}$ and $\mathcal{C}_{X/}$.
(c) The category of *pointed anima* is the slice category $\mathrm{An}_* = \mathrm{An}_{*/}$.
(d) The category of *spectra* is

$$\mathrm{Sp} = \lim\big(\cdots \xrightarrow{\Omega} \mathrm{An}_* \xrightarrow{\Omega} \mathrm{An}_* \xrightarrow{\Omega} \cdots\big),$$

where $\Omega\colon \mathrm{An}_* \to \mathrm{An}_*$ is the loop space functor.[11] Therefore, a spectrum is a family of pointed anima $(X_n)_{n\in\mathbf{Z}}$ with equivalences $\Omega X_{n+1} \simeq X_n$.

*Remark* 2.27 (Spectra as reduced excisive functors). Equivalently, Sp is the full subcategory of $\mathrm{Fun}(\mathrm{An}_*, \mathrm{An})$ consisting of functors that preserve filtered colimits, send $*$ to $*$, and carry pushout squares to pullback squares.[12]

**Definition 2.28** (Presheaf category)**.** Let $\mathcal{C}$ be a small category. Its *presheaf category* is

$$\mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{An}).$$

Its objects are called *presheaves* on $\mathcal{C}$. The *Yoneda embedding* is the functor

$$\text{よ}_{\mathcal{C}}\colon \mathcal{C} \longrightarrow \mathcal{P}(\mathcal{C}), \qquad X \longmapsto \mathrm{Map}_{\mathcal{C}}(-, X).$$

**Theorem 2.29** (Yoneda lemma)**.** *Let $\mathcal{C}$ be a small category. For $X \in \mathcal{C}$ and $F \in \mathcal{P}(\mathcal{C})$, evaluation at $\mathrm{id}_X$ induces a natural equivalence*

$$\mathrm{Map}(\text{よ}_{\mathcal{C}}(X), F) \xrightarrow{\sim} F(X), \qquad f \longmapsto f_X(\mathrm{id}_X).$$

*In particular, the Yoneda embedding is fully faithful.*

**Proposition 2.30.** *Let $\mathcal{C}$ be a small category and $\mathcal{D}$ a category with colimits. Precomposition with the Yoneda embedding induces an equivalence*

$$\text{よ}^*_{\mathcal{C}}\colon \mathrm{Fun}^{\mathrm{colim}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

*Moreover, every presheaf is canonically a colimit of representable presheaves:*

$$F \simeq \operatorname*{colim}_{\text{よ}_{\mathcal{C}}(X)\to F} \text{よ}_{\mathcal{C}}(X).$$

[11] For a pointed anima $(X, x)$, the loop anima is given by the pullback $* \times_X *$.
[12] This description may be more intuitive: it refines the notion of a homology theory.

*Example* 2.31. For $\mathcal{C} = *$, we have $\mathcal{P}(\mathcal{C}) \simeq \mathrm{An}$, and the Yoneda embedding sends $*$ to $*$. In this case, Proposition 2.30 implies that the precomposition with $*\colon * \to \mathrm{An}$ induces an equivalence

$$\mathrm{Fun}^{\mathrm{colim}}(\mathrm{An}, \mathcal{D}) \xrightarrow{\sim} \mathcal{D}.$$

Moreover, every anima $A$ is canonically the colimit $A \simeq \mathrm{colim}_{*\to A} *$.

**Construction 2.32** (Animation)**.** Let $\mathcal{C}$ be a small category with finite coproducts. We denote by $\mathcal{P}_\Sigma(\mathcal{C}) \subseteq \mathcal{P}(\mathcal{C})$ the full subcategory of presheaves carrying finite coproducts in $\mathcal{C}$ to products in An.

*Remark* 2.33. The motivation for Construction 2.32 is that many ordinary categories can be written as $\mathcal{P}_\Sigma(\mathcal{C})_{\leq 0}$, i.e., the subcategory of *set-valued presheaves*. For example,

(a) $\mathrm{Set} \simeq \mathcal{P}_\Sigma(\mathrm{Set}^{\mathrm{fin}})_{\leq 0}$;
(b) $\mathrm{Grp} \simeq \mathcal{P}_\Sigma(\mathrm{Grp}^{\mathrm{fg,free}})_{\leq 0}$;
(c) $\mathrm{CAlg} \simeq \mathcal{P}_\Sigma(\mathrm{CAlg}^{\mathrm{fg,free}})_{\leq 0}$;
(d) $\mathrm{Mod}_R \simeq \mathcal{P}_\Sigma(\mathrm{Mod}_R^{\mathrm{fg,free}})_{\leq 0}$,

where $R$ is an ordinary commutative ring. Removing the truncation gives the corresponding *animations*:

(a) *animated sets* are animae, i.e., $\mathrm{An} \simeq \mathcal{P}_\Sigma(\mathrm{Set}^{\mathrm{fin}})$;
(b) *animated groups* are $\mathbf{E}_1$ groups in An, i.e., $\mathcal{P}_\Sigma(\mathrm{Grp}^{\mathrm{fg,free}})$;
(c) *animated associative algebra* are *connective associative $H\mathbf{Z}$-algebra spectra*, i.e., $\mathcal{P}_\Sigma(\mathrm{Alg}^{\mathrm{fg,free}})$;
(d) $\mathcal{P}_\Sigma(\mathrm{Mod}_R^{\mathrm{fg,free}}) \simeq \mathrm{D}_{\geq 0}(R)$ is the *connective derived category* of $R$-modules.

**Notation 2.34.** Let $\mathcal{C}$ be a category and $S$ a collection of morphisms in $\mathcal{C}$. We denote by

$$\mathrm{L}_S\mathcal{C} = \left\{ x \in \mathcal{C} \;\middle|\; \mathrm{Map}(z, x) \xrightarrow{\sim} \mathrm{Map}(y, x) \text{ for every map } y \to z \in S \right\}$$

the full subcategory of *$S$-local objects*.

**Definition 2.35** (Presentable category)**.** A category is *presentable* if it is equivalent to $\mathrm{L}_S\mathcal{P}(\mathcal{C})$ for a small category $\mathcal{C}$ and a small collection of morphisms $S$ in $\mathcal{P}(\mathcal{C})$.

**Theorem 2.36** (Characterizations of presentability)**.** *Let $\mathcal{C}$ be a category. The following are equivalent:*

*(a) $\mathcal{C}$ is presentable;*
*(b) $\mathcal{C}$ is accessible and admits small limits;*
*(c) $\mathcal{C}$ is accessible and admits small colimits.*

*Example* 2.37. The following categories are presentable:

(a) the category of anima An;
(b) the category of small categories Cat;
(c) the category of $n$-truncated anima $\mathrm{An}_{\leq n}$;
(d) the category of spectra Sp;
(e) the category of groups Grp;
(f) the presheaf category $\mathcal{P}(\mathcal{C})$, for $\mathcal{C}$ small;
(g) $\mathcal{P}_\Sigma(\mathcal{C})$, for $\mathcal{C}$ small with finite coproducts.[13]

The following are not presentable:

(a) the opposite category of anima $\mathrm{An}^{\mathrm{op}}$;
(b) the category of topological spaces Top.

**Theorem 2.38** (Bousfield localization)**.** *Let $\mathcal{C}$ be a presentable category and $S$ a small collection of morphisms in $\mathcal{C}$. The full subcategory $\mathrm{L}_S\mathcal{C}$ of $S$-local objects is presentable, and its inclusion admits a left adjoint $\mathrm{L}_S\colon \mathcal{C} \longrightarrow \mathrm{L}_S\mathcal{C}$. Thus, it is a left Bousfield localization.*

[13] Take $S$ to consist of the maps $\text{よ}_{\mathcal{C}}(x_1) \sqcup \cdots \sqcup \text{よ}_{\mathcal{C}}(x_n) \to \text{よ}_{\mathcal{C}}(x_1 \sqcup \cdots \sqcup x_n)$ for all finite families, including the empty family.

**Theorem 2.39** (Adjoint functor theorem)**.** *Let $f\colon \mathcal{C} \to \mathcal{D}$ be a functor between locally small categories.*

(a) *If $\mathcal{C}$ is presentable, then $f$ admits a right adjoint if and only if it preserves small colimits.*
(b) *If $\mathcal{C}$ and $\mathcal{D}$ are presentable, then $f$ admits a left adjoint if and only if it is accessible and preserves small limits.*

**Notation 2.40.**

(a) $\mathrm{Pr}^L \subseteq \widehat{\mathrm{Cat}}$ is the category of presentable categories and left adjoint functors;
(b) $\mathrm{Pr}^R \subseteq \widehat{\mathrm{Cat}}$ is the category of presentable categories and right adjoint functors.

Limits of categories are relatively easy to describe, whereas colimits are more difficult. For presentable categories, both limits and colimits can be computed using limits of underlying categories.

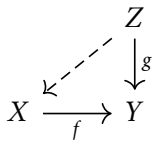
**Proposition 2.41** (Closure properties)**.** *The inclusions $\mathrm{Pr}^L \hookrightarrow \widehat{\mathrm{Cat}}$ and $\mathrm{Pr}^R \hookrightarrow \widehat{\mathrm{Cat}}$ preserve small limits.*

**Corollary 2.42.** *The category $\mathrm{Pr}^L$ admits small limits and colimits.*[14]

**Proposition 2.43.** *Let $\mathcal{C}$ and $\mathcal{D}$ be presentable categories. The functor categories $\mathrm{Fun}^{\mathrm{colim}}(\mathcal{C},\mathcal{D})$ and $\mathrm{Fun}^{\lim}(\mathcal{C}^{\mathrm{op}},\mathcal{D})$ are presentable.*

**Definition 2.44** ($n$-truncated objects and morphisms)**.** Let $\mathcal{C}$ be a category and $n \geq -2$ an integer.

(a) An object $X \in \mathcal{C}$ is *$n$-truncated* if $\mathrm{Map}_{\mathcal{C}}(Z,X)$ is $n$-truncated for every $Z \in \mathcal{C}$. We denote the full subcategory of such objects by $\mathcal{C}_{\leq n} \hookrightarrow \mathcal{C}$. When its inclusion admits a left adjoint, we denote it by $\tau_{\leq n}$; this holds, for example, if $\mathcal{C}$ is presentable.
(b) A morphism $f\colon X \to Y$ is *$n$-truncated* if it is an $n$-truncated object of $\mathcal{C}_{/Y}$. Equivalently, for every morphism $g\colon Z \to Y$, the anima of lifts in the diagram




is $n$-truncated.

*Example* 2.45 ($n$-truncated morphisms)*.*

(a) The $(-2)$-truncated morphisms are the equivalences.
(b) The $(-1)$-truncated morphisms are the monomorphisms.

*Exercise* 2.46*.* Show the following:

(a) A map of anima is $n$-truncated if and only if its fibers are $n$-truncated.
(b) If $\mathcal{C}$ admits pullbacks, truncatedness of a morphism $f\colon X \to Y$ can be characterized inductively:
- $f$ is $(-2)$-truncated if and only if it is an equivalence;
- for $n \geq -1$, $f$ is $n$-truncated if and only if the diagonal $\Delta_f\colon X \to X \times_Y X$ is $(n-1)$-truncated.

**Definition 2.47** (Sieve)**.** Let $\mathcal{C}$ be a small category and $X \in \mathcal{C}$. A *sieve* on $X$ is a subpresheaf of $\text{よ}_{\mathcal{C}}(X) = \mathrm{Map}_{\mathcal{C}}(-,X)$, i.e., a monomorphism $R \hookrightarrow \text{よ}_{\mathcal{C}}(X)$ in $\mathcal{P}(\mathcal{C})$.

**Definition 2.48** (Grothendieck topology)**.** A *Grothendieck topology* or a *topology* $\tau$ on a small category $\mathcal{C}$ is a collection of sieves, called *$\tau$-covering sieves*, satisfying the following conditions:

(a) If $R \hookrightarrow \text{よ}_{\mathcal{C}}(X)$ is $\tau$-covering and $f\colon Y \to X$, then

$$f^*R = \left(R \times_{\text{よ}_{\mathcal{C}}(X)} \text{よ}_{\mathcal{C}}(Y)\right) \hookrightarrow \text{よ}_{\mathcal{C}}(Y)$$

is $\tau$-covering.
(b) The maximal sieve $\mathrm{id}_{\text{よ}_{\mathcal{C}}(X)}\colon \text{よ}_{\mathcal{C}}(X) \to \text{よ}_{\mathcal{C}}(X)$ is $\tau$-covering for every $X$.

[14] For colimits, use the equivalence $(\mathrm{Pr}^L)^{\mathrm{op}} \simeq \mathrm{Pr}^R$.

(c) Let $R, S \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$ be sieves. If $R$ is $\tau$-covering and $f^*S$ is $\tau$-covering for every $f\colon Y \to X$ in $R$, then $S$ is $\tau$-covering.

*Exercise* 2.49. Let $\tau$ be a topology on a small category $\mathcal{C}$. Show the following:

(a) If $R \hookrightarrow S \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$ and $R$ is $\tau$-covering, then $S$ is $\tau$-covering.
(b) If $R, S \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$ are $\tau$-covering, then so is the intersection

$$R \cap S = R \times_{\text{ﻉ}_{\mathcal{C}}(X)} S \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X).$$

*Remark* 2.50 (Topologies generated by sieves). Any intersection $\bigcap_{i\in I} \tau_i$ of topologies on a small category $\mathcal{C}$ is a topology. In particular, any collection of sieves generates a topology.

*Example* 2.51. Let $\mathrm{L}\colon \mathcal{P}(\mathcal{C}) \to \mathcal{T}$ be a left exact[15] left Bousfield localization. Declare a sieve $R \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$ to be covering if, after applying L, the map $\mathrm{L}(R) \to \mathrm{L}(\text{ﻉ}_{\mathcal{C}}(X))$ is an equivalence. This defines a Grothendieck topology on $\mathcal{C}$.

**Definition 2.52** (Sheaf)**.** Let $\tau$ be a topology on a small category $\mathcal{C}$. A *$\tau$-sheaf* is a presheaf $F \in \mathcal{P}(\mathcal{C})$ such that, for every $\tau$-covering sieve $R \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$, the restriction map

$$F(X) \simeq \mathrm{Map}(\text{ﻉ}_{\mathcal{C}}(X), F) \longrightarrow \mathrm{Map}(R, F)$$

is an equivalence. The full subcategory of $\tau$-sheaves is the left Bousfield localization $\mathrm{L}_\tau\mathcal{P}(\mathcal{C}) \subseteq \mathcal{P}(\mathcal{C})$.

**Theorem 2.53** (Lurie)**.** *Let $\tau$ be a Grothendieck topology on a small category $\mathcal{C}$.*

*(a) The left Bousfield localization $\mathrm{L}_\tau\colon \mathcal{P}(\mathcal{C}) \to \mathrm{L}_\tau\mathcal{P}(\mathcal{C})$ is left exact.*
*(b) A sieve $R \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)$ is $\tau$-covering if and only if $\mathrm{L}_\tau(R) \to \mathrm{L}_\tau(\text{ﻉ}_{\mathcal{C}}(X))$ is an equivalence.*
*(c) The assignment $\tau \mapsto \mathrm{L}_\tau$ gives an injection*

$$\{\textit{Grothendieck topologies on } \mathcal{C}\} \hookrightarrow \{\textit{left exact localizations of } \mathcal{P}(\mathcal{C})\}$$

*Proof sketch.* Sheafification can be constructed explicitly using the *plus construction*. For $F \in \mathcal{P}(\mathcal{C})$, set

$$F^+(X) = \operatorname*{colim}_{R \hookrightarrow \text{ﻉ}_{\mathcal{C}}(X)} \mathrm{Map}(R, F),$$

where the colimit ranges over $\tau$-covering sieves ordered by reverse inclusion. This poset is filtered, so $(-)^+$ preserves finite limits. Furthermore, for every $\tau$-sheaf $G$, the restriction $F \to F^+$ induces an equivalence

$$\mathrm{Map}(F^+, G) \xrightarrow{\sim} \mathrm{Map}(F, G).$$

If $F$ is set-valued, then $(F^+)^+$ is a $\tau$-sheaf,[16] and hence by the above equivalence, $\mathrm{L}_\tau F \simeq (F^+)^+$. More generally, if $F$ is $n$-truncated, for a finite integer $n \geq -2$, then $\mathrm{L}_\tau F \simeq F^{+(n+2)}$. In general, the difficulty is that $\mathrm{Map}(R, -)$ need not preserve filtered colimits. Choose a sufficiently large regular cardinal $\kappa$ such that these functors preserve $\kappa$-filtered colimits for every covering sieve $R$. In general, transfinite iteration gives $\mathrm{L}_\tau F \simeq F^{+(\kappa)}$. Afterwards, (a), (b) and (c) follows immediately. □

**Theorem 2.54** (Characterization of topoi)**.** *Let $T$ be a presentable category. The following are equivalent:*

*(a) There are a small category $\mathcal{C}$ and a small collection $S$ of morphisms in $\mathcal{P}(\mathcal{C})$ such that $T \simeq \mathrm{L}_S\mathcal{P}(\mathcal{C})$ and the localization $\mathrm{L}_S\colon \mathcal{P}(\mathcal{C}) \to T$ is left exact.*
*(b) The* self-indexing functor

$$T^{\mathrm{op}} \longrightarrow \widehat{\mathrm{Cat}}, \quad X \longmapsto T_{/X},$$

*with transition functors given by pullback, preserves small limits.*

*In this case, $T$ is called a* topos*.*

[15] i.e., finite limit-preserving
[16] In the classical literature, this is a messy computation.

*Remark* 2.55. Condition (b) of Theorem 2.54 has the following heuristic interpretation. For a presentable category $\mathcal{C}$, the adjoint functor theorem gives an equivalence

$$\text{よ}_{\mathcal{C}}\colon \mathcal{C} \xrightarrow{\sim} \mathrm{Fun}^{\lim}(\mathcal{C}^{\mathrm{op}}, \mathrm{An}), \qquad x \longmapsto \mathrm{Map}_{\mathcal{C}}(-, x),$$

induced by the Yoneda embedding. Since the groupoid core functor $(-)^{\simeq}$ preserves limits, $T^{\mathrm{op}} \to \widehat{\mathrm{Cat}}$ preserves limits suggests that, modulo size issues, $T$ exists inside itself.

However, the collection of sets do not form a set, and the collection of groupoids, do not form a groupoid. The self-indexing property illustrates why $(\infty, 1)$-category theory is better behaved than ordinary category theory, see Exercise 2.58.

**Corollary 2.56.** *For every anima $A$, there is an equivalence*

$$\mathrm{An}_{/A} \xrightarrow{\sim} \mathrm{Fun}(A, \mathrm{An}), \quad (X \to A) \mapsto (a \mapsto X_a),$$

*whose inverse sends $F$ to the map* $\mathrm{colim}_{a\in A} F(a) \to \mathrm{colim}_{a\in A} * \simeq A$.

*Proof.* Since An is a topos, Theorem 2.54 implies that its self-indexing functor preserves small limits, hence

$$\mathrm{An}_{/A} \simeq \lim_{a\in A} \mathrm{An}_{/*} \simeq \lim_{a\in A} \mathrm{Fun}(*, \mathrm{An}) \simeq \mathrm{Fun}(A, \mathrm{An}),$$

using $A \simeq \mathrm{colim}_{a\in A} *$. □

*Remark* 2.57. For a set $S$, Corollary 2.56 gives the elementary equivalence $\mathrm{Set}_{/S} \simeq \mathrm{Fun}(S, \mathrm{Set})$.

*Exercise* 2.58.

(a) Show that Set is not a topos.[17]
(b) Show that $\mathrm{An}_{\leq n}$ is not a topos for $n \geq -1$.

**Definition 2.59** (Čech nerve)**.** Let $\mathcal{C}$ be a category with pullbacks and $f\colon U \to X$ a morphism. The *Čech nerve* of $f$ is the simplicial object $\check{\mathrm{C}}_\bullet(f)\colon \Delta^{\mathrm{op}} \to \mathcal{C}_{/X}$ whose diagram has the form

$$\cdots \overset{\to}{\to} U \times_X U \rightrightarrows U \to X.$$

More precisely, it is the right Kan extension of $f$ along the inclusion $\Delta^{\mathrm{op}}_{\leq 0} \hookrightarrow \Delta^{\mathrm{op}}$:

$$\begin{array}{ccc} \Delta^{\mathrm{op}}_{\leq 0} & \xrightarrow{f} & \mathcal{C}_{/X}, \\ \downarrow & \nearrow_{\check{\mathrm{C}}_\bullet(f)} & \\ \Delta^{\mathrm{op}} & & \end{array}$$

where $\Delta^{\mathrm{op}}_{\leq 0} \simeq *$.

**Definition 2.60** (Effective epimorphism)**.** Let $\mathcal{C}$ be a category with pullbacks. A morphism $f\colon U \to X$ is an *effective epimorphism* if the geometric realization of $\check{\mathrm{C}}_\bullet(f)$ exists and the canonical map

$$\operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} \check{\mathrm{C}}_n(f) \to X$$

is an equivalence.

*Example* 2.61.

(a) In Set, Grp, CAlg and similar algebraic categories,[18] the colimit of the Čech nerve is the coequalizer

$$\operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} \check{\mathrm{C}}_n(f) \simeq \mathrm{coeq}(U \times_X U \rightrightarrows U) \simeq \mathrm{im}(f),$$

by explicit computations. Therefore, $f$ is an effective epimorphism if and only if it is surjective.

[17] Indeed, try to show that the condition (b) of Theorem 2.54 fails.
[18] Indeed, the forgetful functor to Set preserves geometric realization and reflects isomorphism.

(b) In An, a morphism $f\colon U \to X$ is an effective epimorphism if and only if

$$\tau_{\leq 0}(f)\colon \tau_{\leq 0}(U) \longrightarrow \tau_{\leq 0}(X)$$

is surjective, i.e., if $f$ is essentially surjective as a functor.

*Remark* 2.62. In ordinary category theory, every effective epimorphism is an epimorphism. This already fails for $(2,1)$-categories. For example, an epimorphism of animae induces an isomorphism on 0-truncations,[19] whereas the effective epimorphism $* \sqcup * \to *$ is not an epimorphism in An, or even in $\mathrm{An}_{\leq 1}$. Indeed, $* \sqcup * \to *$ is an epimorphism in $\mathrm{An}_{\leq 1}$ if and only if, for every $X \in \mathrm{An}_{\leq 1}$, the precomposition map

$$X \simeq \mathrm{Map}_{\mathrm{An}}(*, X) \to \mathrm{Map}_{\mathrm{An}}(* \sqcup *, X) \simeq X \times X$$

is a monomorphism, or equivalently, $\Delta_X\colon X \to X \times X$ is $(-1)$-truncated, i.e., $X$ is 0-truncated.

**Definition 2.63** (Factorization system)**.** Let $\mathcal{C}$ be a category. A *factorization system* $(\mathcal{C}, L, R)$ consists of two classes of morphisms $L$ and $R$ satisfying the following conditions:

(a) $L$ and $R$ contain all equivalences and are closed under composition;
(b) every morphism $f$ factors uniquely as $f \simeq r \circ l$, with $l \in L$ and $r \in R$.

*Example* 2.64. In Set, Grp, CAlg, and similar algebraic categories, surjections, i.e., effective epimorphisms, and injections, i.e., monomorphisms, form a factorization system.

*Remark* 2.65 (Factorization system on An and Cat)*.*

(a) Let $T$ be a topos. The effective epimorphisms and monomorphisms form a factorization system on $T$.
(b) The essentially surjective functors and fully faithful functors form a factorization system on Cat.

**Definition 2.66** ($n$-connectedness)**.** Let $T$ be a topos and $n \geq -2$.

(a) $X \in T$ is *$n$-connected* if $\tau_{\leq n}(X) \simeq *$;
(b) $f\colon Y \to X$ is *$n$-connected* if it is an $n$-connected object of the slice topos $T_{/X}$.

*Remark* 2.67. By the equivalence $\mathrm{An}_{/X} \simeq \mathrm{Fun}(X, \mathrm{An})$, a map of animae is $n$-connected if and only if all its fibers are $n$-connected. Indeed, $n$-truncation in the functor category is computed pointwise.

**Proposition 2.68.** *Let $T$ be a topos.*

*(a) Every morphism in $T$ is $(-2)$-connected.*
*(b) A morphism $f\colon Y \to X$ in $T$ is $(-1)$-truncated if and only if $f$ is an effective epimorphism.*

*Proof.* The morphism $f$ is $(-1)$-connected if and only if $\tau_{\leq -1}(f) \simeq \mathrm{id}_X$. Since $\tau_{\leq -1}$ is left adjoint to the inclusion $(T_{/X})_{\leq -1} \hookrightarrow T_{/X}$, the map $\tau_{\leq -1}(f)$ is the smallest subobject of $X$ through which $f$ factors. By Remark 2.65, $f$ can be factored as $Y \to X' \hookrightarrow X$, where $Y \to X'$ is an effective epimorphism. If $f$ is $(-1)$-connected, then $X' \simeq X$, so $f$ is an effective epimorphism.

Conversely, suppose that $f$ is an effective epimorphism and factors through a subobject $X' \hookrightarrow X$, where $Y \to X'$ is an effective epimorphism. Moreover, factor $Y \to X'$ as an effective epimorphism $Y \to X''$ followed by a monomorphism $X'' \hookrightarrow X'$. By the uniqueness of the epi-mono factorization of $f$, there exists an isomorphism $X'' \simeq X$ over $X$. Since $(T_{/X})_{\leq -1}$ is a poset and $X'' \subseteq X' \subseteq X$, we obtain $X' \simeq X$. Thus, $f$ is $(-1)$-connected. □

*Remark* 2.69. Let $T$ be a topos.

(a) For every $n \geq -2$, the $n$-connected morphisms and the $n$-truncated morphisms form a factorization system on $T$.

[19] A morphism of animae $f\colon X \to Y$ is an epimorphism if and only if it induces an isomorphism on homology with local abelian coefficients. In particular, taking the 0-th homology with constant coefficients $\mathbf{Z}$ gives us the desired result.

(b) For $n \geq -1$, a morphism $Y \to X$ in $T$ is $n$-connected if and only if it is an effective epimorphism and its diagonal $Y \to Y \times_X Y$ is $(n-1)$-connected.

*Example* 2.70 (Universal covering). For $n = 0$, a pointed object $x : * \to X$ admits a factorization $* \to \widetilde{X} \to X$, where the first map is 0-connected and the second is 0-truncated. This is the *universal covering* of the component of $X$ containing the base point.

*Remark* 2.71 (Descent via Čech nerves). Let $\mathcal{C}$ be a small category with pullback and $\mathcal{U} = (U_i \to X)_{i \in I}$ be a family of morphisms in $\mathcal{C}$, where $I$ is a set. Consider the epi-mono factorization in $\mathcal{P}(\mathcal{C})$:

$$\coprod_{i \in I} \text{ю}_{\mathcal{C}}(U_i) \to \langle \mathcal{U} \rangle \hookrightarrow \text{ю}_{\mathcal{C}}(X),$$

This implies that $\langle \mathcal{U} \rangle \subseteq \text{ю}_{\mathcal{C}}(X)$ is the sieve generated by $\mathcal{U}$, i.e., it consists of the morphisms $f : Y \to X$ that factor through some $U_i$. By the epi-mono factorization,

$$\begin{aligned}
\langle \mathcal{U} \rangle &= \operatorname*{colim}_{[n] \in \Delta^{\mathrm{op}}} \check{\mathrm{C}}_n \left( \coprod_{i \in I} \text{ю}_{\mathcal{C}}(U_i) \to \text{ю}_{\mathcal{C}}(X) \right) \\
&= \operatorname*{colim}_{[n] \in \Delta^{\mathrm{op}}} \left( \coprod_{i_0, \dots, i_n} \left( \text{ю}_{\mathcal{C}} \left( U_{i_0} \times_{\text{ю}_{\mathcal{C}}(X)} \dots \times_{\text{ю}_{\mathcal{C}}(X)} \text{ю}_{\mathcal{C}}(U_{i_n}) \right) \right) \right) \\
&= \operatorname*{colim}_{[n] \in \Delta^{\mathrm{op}}} \left( \coprod_{i_0, \dots, i_n} \left( \text{ю}_{\mathcal{C}}(U_{i_0} \times_X \dots \times_X U_{i_n}) \right) \right).
\end{aligned}$$

Therefore, $F \in \mathcal{P}(\mathcal{C})$ satisfies descent with respect to the sieve $\langle \mathcal{U} \rangle \hookrightarrow \text{ю}_{\mathcal{C}}(X)$, if and only if the map

$$F(X) \to \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(\langle \mathcal{U} \rangle, F) \simeq \lim_{[n] \in \Delta} \prod_{i_0, \dots, i_n \in I} F(U_{i_0} \times_X \dots \times_X U_{i_n}).$$

Equivalently, the descent condition says that the canonical map

$$F(X) \to \lim \left( \prod_{i \in I} F(U_i) \rightrightarrows \prod_{i,j \in I} F(U_i \times_X U_j) \mathrel{\substack{\rightarrow \\ \rightarrow}} \dots \right)$$

is an isomorphism.

**Algebraic geometry.** The goal of this section is to prove the *tubular neighborhood theorem*.

**Theorem 2.72** (Tubular neighborhood theorem)**.** *Let $X$ be a smooth manifold and $Z \hookrightarrow X$ a closed smooth submanifold. There is a zigzag of open embeddings*

$$\begin{array}{ccccc}
Z & = & Z & = & Z \\
\downarrow & & \downarrow & & \downarrow \\
X & \longleftarrow & U & \longrightarrow & \mathcal{N}_{Z/X},
\end{array}$$

*where $\mathcal{N}_{Z/X}$ is the normal bundle, both maps out of $U$ are open embeddings, and $Z \hookrightarrow \mathcal{N}_{Z/X}$ is the zero section.*

In algebraic geometry, we consider smooth affine schemes instead. This parallels the reconstruction of smooth manifolds as $\mathrm{C}^\infty$-spectra of their $\mathrm{C}^\infty$-rings of smooth functions.

**Theorem 2.73** (Algebraic tubular neighborhood theorem)**.** *Consider a commutative diagram of affine schemes*

$$\begin{array}{ccc}
Z & \hookrightarrow & X \\
& \searrow & \downarrow \\
& & S,
\end{array}$$

*where $Z \hookrightarrow X$ is a closed immersion and both $Z$ and $X$ are smooth over $S$. Then there is a zigzag*

$$\begin{array}{ccccc} Z & = & Z & = & Z \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & V & \longrightarrow & \mathbf{V}(\mathcal{N}_{Z/X}), \end{array}$$

*in which $Z \hookrightarrow \mathbf{V}(\mathcal{N}_{Z/X})$ is the zero section, the maps $V \to X$ and $V \to \mathcal{N}_{Z/X}$ are étale, and both squares are cartesian.*

**Theorem 2.74** (Characterization of smoothness)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite presentation. The following conditions are equivalent:*

(a) *$f$ is formally smooth, i.e., for every ring $A$, square-zero ideal $I \subseteq A$, and commutative square*

$$\begin{array}{ccc} \operatorname{Spec}(A/I) & \longrightarrow & Y \\ \downarrow & \nearrow & \downarrow \\ \operatorname{Spec}(A) & \longrightarrow & X, \end{array}$$

*there is a dashed lift making both triangles commute.*

(b) *The cotangent complex $\mathrm{L}_f$ is a locally free sheaf on $Y$.*[20]

(c) *$f$ is flat and $\Omega_f$ is locally free of rank $y \mapsto \dim_y(Y_{f(y)})$.*

(d) *For every $x\colon \operatorname{Spec}(k) \to X$ with $k$ algebraically closed, the fiber $Y_x$ is a regular scheme.*

(e) *For every $y \in Y$, there are affine open neighborhoods $y \in V$ and $f(y) \in U$, with $f(V) \subseteq U$, and factorization*

$$\begin{array}{ccc} V & \hookrightarrow & \mathbf{A}^n \times U \\ & \searrow & \downarrow \mathrm{pr}_U \\ & & U, \end{array}$$

*where the closed immersion $V \hookrightarrow \mathbf{A}^n \times U$ is defined by $f_1, \dots, f_m \in \mathcal{O}(U)[x_1, \dots, x_n]$ and the Jacobian*

$$\left(\frac{\partial f_i}{\partial x_j}\right)_{\substack{1 \le i \le m \\ 1 \le j \le n}}$$

*has rank $m$ at every point of $V$.*

*Remark* 2.75.

(a) Let $A = R[\epsilon]/(\epsilon^2)$ be the ring of *dual numbers*, and let $I = (\epsilon)$, so that $A/I \simeq R$. Condition (a) of Theorem 2.74 says that, for every $R$-point $y\colon \operatorname{Spec}(R) \to Y$ and tangent vector $v\colon \operatorname{Spec}(A) \to X$ at $f(y)$, there is a lift of $v$ to $Y$ restricting to $y$. In other words, $f$ is surjective on tangent spaces.

The general lifting condition is stronger. For example, applying it successively to

$$R[\epsilon]/(\epsilon^{n+1}) \twoheadrightarrow R[\epsilon]/(\epsilon^n) \quad \text{and} \quad I_n = (\epsilon^n)/(\epsilon^{n+1}),$$

where $n \ge 1$, shows that *jets* lift as well.

(b) Condition (d) of Theorem 2.74 means that, for every geometric point $x$ of $X$ and every $y \in Y_x$, the stalk $\mathcal{O}_{Y_x,y}$ is a regular local ring, i.e.,

$$\dim_{\kappa(y)}(\mathfrak{m}_y/\mathfrak{m}_y^2) = \dim(\mathcal{O}_{Y_x,y}).$$

[20] The map $f$ is locally of finite presentation ensures that $L_f$ is finitely generated.

*Example* 2.76. The morphism $\mathbf{A}^n_R \to \mathrm{Spec}(R)$ is smooth. Its infinitesimal lifting problem translates into

$$\begin{array}{ccc} R & \longrightarrow & A \\ \downarrow & \nearrow & \downarrow \\ R[x_1, \ldots, x_n] & \longrightarrow & A/I. \end{array}$$

**Theorem 2.77** (Characterization of étaleness)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite presentation. The following conditions are equivalent to $f$ being étale:*

(a) *$f$ is formally étale: for every ring $A$, square-zero ideal $I \subseteq A$, and commutative square*

$$\begin{array}{ccc} \mathrm{Spec}(A/I) & \longrightarrow & Y \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & X, \end{array}$$

*there is a unique dashed lift making both triangles commute.*

(b) *The cotangent complex $\mathrm{L}_f$ vanishes.*

(c) *$f$ is flat and $\Omega_f = 0$.*

(d) *$f$ is flat and, for every algebraically closed field $k$ and every $x\colon \mathrm{Spec}(k) \to X$, the fiber $Y_x$ is a coproduct of copies of $\mathrm{Spec}(k)$.*

(e) *For every $y \in Y$, there are affine open neighborhoods $y \in V$ and $f(y) \in U$, with $f(V) \subseteq U$, and factorization*

$$\begin{array}{ccc} V & \hookrightarrow & \mathbf{A}^n \times U \\ & \searrow & \downarrow \mathrm{pr}_U \\ & & U, \end{array}$$

*where the closed immersion $V \hookrightarrow \mathbf{A}^n \times U$ is defined by $f_1, \ldots, f_n \in \mathcal{O}(U)[x_1, \ldots, x_n]$ and the Jacobian*

$$\left(\frac{\partial f_i}{\partial x_j}\right)_{1 \le i,j \le n}$$

*has rank $n$ at every point of $V$.*

**Theorem 2.78** (Local structure of smooth morphisms)**.** *Let $f\colon Y \to X$ be a morphism of schemes locally of finite presentation. Then $f$ is smooth if and only if, for every $y \in Y$, there are affine open neighborhoods $y \in V \subseteq Y$ and $f(y) \in U \subseteq X$, with $f(V) \subseteq U$, and a factorization*

$$\begin{array}{ccc} V & \longrightarrow & \mathbf{A}^n \times U \\ & \searrow^{f|_V} & \downarrow \mathrm{pr}_U \\ & & U, \end{array}$$

*for some $n \ge 0$, in which the map $V \to \mathbf{A}^n \times U$ is étale.*

*Exercise* 2.79. Let $k'/k$ be a finite field extension. Show that it is smooth if and only if it is separable, using the different characterizations of smoothness and étaleness.

**Notation 2.80.** Let $X$ be a scheme.

(a) $\mathrm{Sch}_X = \mathrm{Sch}_{/X}$ denotes the category of schemes over $X$.

(b) $\mathrm{Sm}_X \subseteq \mathrm{Sch}_X$ denotes the full subcategory of smooth $X$-schemes.

(c) $\mathrm{Et}_X \subseteq \mathrm{Sm}_X$ denotes the full subcategory of étale $X$-schemes.

The following observation motivates the use of smooth schemes in motivic homotopy theory.

*Remark* 2.81. The category $\mathrm{Sm}_X$ is the smallest full subcategory $\mathcal{C} \subseteq \mathrm{Sch}_X$ containing $X$ such that

(a) for every $Y \in \mathrm{Sch}_X$ and open cover $(V_i \subseteq Y)_{i\in I}$, $V_i \in \mathcal{C}$ for every $i$ implies that $Y \in \mathcal{C}$;
(b) if $Y \in \mathcal{C}$ and $Y' \to Y$ is étale, then $Y' \in \mathcal{C}$;
(c) if $Y \in \mathcal{C}$, then $\mathbf{A}^1 \times Y \in \mathcal{C}$.

Conditions (b) and (c) provide the setting for *homotopy-invariant sheaves* on $\mathcal{C}$.

**Definition 2.82** (Derivations and differentials)**.** Let $A \to B$ be a map of commutative rings and $M$ a $B$-module.

(a) The *split square-zero extension* $B \oplus M \in \mathrm{CAlg}$ has componentwise addition and multiplication
$$(b, m)(b', m') = (bb', bm' + b'm).$$
(b) An $A$-*derivation* of $B$ into $M$ is a section of $B \oplus M \twoheadrightarrow B$ in $\mathrm{CAlg}_A$, i.e., a map $\mathrm{d}\colon B \to M$ satisfying
$$\mathrm{d}(x + y) = \mathrm{d}x + \mathrm{d}y, \qquad \mathrm{d}|_A = 0, \qquad \mathrm{d}(xy) = x\mathrm{d}y + y\mathrm{d}x.$$
(c) The set of $A$-derivations is
$$\mathrm{Der}_A(B, M) = \mathrm{Map}_{\mathrm{CAlg}_A/B}(B, B \oplus M).$$
This defines a functor $\mathrm{Der}_A(B, -)\colon \mathrm{Mod}_B \to \mathrm{Set}$ preserving limits and $\kappa$-filtered colimits for every infinite regular cardinal $\kappa > |B|$. Hence it is represented by a $B$-module $\Omega_{B/A}$:
$$\mathrm{Der}_A(B, -) \simeq \mathrm{Map}_{\mathrm{Mod}_B}(\Omega_{B/A}, -).$$
Explicitly,
$$\Omega_{B/A} = \left(\bigoplus_{b\in B} B\mathrm{d}b\right)/\sim,$$
where the relations are those of an $A$-derivation.

*Remark* 2.83 (Cotangent complex)*.*

(a) Let $A \to B$ be a morphism of ordinary rings. The cotangent complex is the animated analogue of the module of differentials. We replace $\mathrm{Mod}_B$ by the connective derived category $\mathrm{D}_{\geq 0}(B)$, i.e., the animation of the category of finitely generated free $B$-modules. Equivalently,
$$\mathrm{D}_{\geq 0}(B) \simeq \mathrm{Ch}_{\geq 0}(\mathrm{Mod}_B)[W^{-1}],$$
where $W$ is the class of quasi-isomorphisms.

Similarly, we replace $\mathrm{CAlg}_A$ by the category $\mathrm{CAlg}_A^{\mathrm{an}}$ of animated commutative $A$-algebras, i,e, the animation of the category of the polynomial algebras $A[x_1, \ldots, x_n]$ for every $n \geq 0$. Equivalently,
$$\mathrm{CAlg}_A^{\mathrm{an}} \simeq \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{CAlg}_A)[W^{-1}],$$
where $W$ is the class of weak equivalences of simplicial algebras.

For $M \in \mathrm{D}_{\geq 0}(B)$, there exists a split square-zero extension $B \oplus M \to B$ in $\mathrm{CAlg}^{\mathrm{an}}$. We again define $\mathrm{Der}_A(B, M)$ to be the anima of $A$-algebra sections of this map. The resulting functor
$$\mathrm{Der}_A(B, -)\colon \mathrm{D}_{\geq 0}(B) \longrightarrow \mathrm{An}$$
is represented by the *cotangent complex* $\mathrm{L}_{B/A} \in \mathrm{D}_{\geq 0}(B)$, i.e.,
$$\mathrm{Der}_A(B, M) \simeq \mathrm{Map}_{\mathrm{D}_{\geq 0}(B)}(\mathrm{L}_{B/A}, M).$$
Since $\mathrm{Mod}_B$ identifies with the full subcategory of 0-truncated objects in $\mathrm{D}_{\geq 0}(B)$, the universal property gives $\tau_{\leq 0}\mathrm{L}_{B/A} \simeq \Omega_{B/A}$.

(b) Similarly, for a morphism of animated rings $C \to D$, there is the category of *connective* $B$-modules $\mathrm{Mod}_B^{\mathrm{cn}}$, which recovers $\mathrm{D}_{\geq 0}(B)$ when $B$ is discrete. The corresponding derivation functor
$$\mathrm{Der}_A(B, -)\colon \mathrm{Mod}_B^{\mathrm{cn}} \to \mathrm{An}$$
is again represented by the cotangent complex $\mathrm{L}_{B/A} \in \mathrm{Mod}_B^{\mathrm{cn}}$.

**Definition 2.84** (Conormal module)**.** Let $A \to B$ be a morphism of animated rings. The *conormal module* of $A \to B$ is $\mathcal{N}_{B/A} = \mathrm{H}_1(\mathrm{L}_{B/A})$.

**Theorem 2.85** (Fundamental exact sequence)**.** *Let $A \to B \to C$ be morphisms of animated rings.*

*(a) There is a fiber sequence*[21] *in* $\mathrm{Mod}_C^{\mathrm{cn}}$*:*

$$\mathrm{L}_{B/A} \otimes_B^{\mathbf{L}} C \longrightarrow \mathrm{L}_{C/A} \longrightarrow \mathrm{L}_{C/B}.$$

*The associated long exact homology sequence contains*

$$\begin{aligned} \cdots \longrightarrow \mathrm{H}_2(\mathrm{L}_{C/B}) \longrightarrow \mathrm{H}_1(\mathrm{L}_{B/A} \otimes_B^{\mathbf{L}} C) \longrightarrow \mathcal{N}_{C/A} \longrightarrow \mathcal{N}_{C/B} \\ \longrightarrow \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0. \end{aligned}$$

*(b) In the ordinary case, if $B \to C$ is smooth, this gives a short exact sequence*

$$0 \longrightarrow \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0.$$

*(c) In the ordinary setting, if instead $A \to C$ is a smooth map and $B \twoheadrightarrow C$ is surjective, we obtain*

$$0 \longrightarrow \mathcal{N}_{C/B} \longrightarrow \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0.$$

*Example* 2.86. Let $A \twoheadrightarrow B$ be a surjection of ordinary rings with kernel $I$. Then $\Omega_{B/A} = 0$ and $\mathcal{N}_{B/A} \simeq I/I^2$ as $B$-modules.

*Remark* 2.87. The analogous exact sequence for a smooth embedding $Z \hookrightarrow X$ of smooth manifolds is

$$0 \longrightarrow \mathrm{T}_Z \longrightarrow \mathrm{T}_X|_Z \longrightarrow \mathcal{N}_{Z/X} \longrightarrow 0,$$

where $\mathcal{N}_{Z/X}$ is the normal bundle.

**Proposition 2.88.** *Consider a cocartesian square of animated rings*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B'. \end{array}$$

*Then there exists a canonical isomorphism* $\mathrm{L}_{B'/A'} \simeq \mathrm{L}_{B/A} \otimes_B^{\mathbf{L}} B'$.

*Remark* 2.89. Let $R$, $S$ and $A$ be ordinary rings, $I \subseteq A$ be a square-zero ideal. The lifting problem

$$\begin{array}{ccc} R & \longrightarrow & A \\ \downarrow & \nearrow & \downarrow \\ S & \longrightarrow & A/I \end{array}$$

can be reformulated using the cartesian square

$$\begin{array}{ccc} \tilde{S} & \longrightarrow & A \\ \downarrow & & \downarrow \\ S & \longrightarrow & A/I. \end{array}$$

Indeed, a lift is equivalent to an $R$-algebra section of $\tilde{S} = S \times_{A/I} A \to S$.

*Remark* 2.90. Let $R \to S$ be a morphism of ordinary rings and $I$ an $S$-module. Square-zero extensions of $S$ by $I$ as an $R$-algebra correspond to animated $R$-derivations of $S$ into $I[1]$. Indeed, consider the extension

$$0 \to I \to \tilde{S} \to S \to 0,$$

[21] It is also a cofiber sequence.

i.e., $\tilde{S} \to S$ is surjective with kernel $I$ such that $I^2 = 0$. By Example 2.86, $\tau_{\leq 1} \mathrm{L}_{\tilde{S}/S} \simeq I[1]$, hence Theorem 2.85(a) supplies a morphism $\mathrm{L}_{S/R} \to I[1]$, which corresponds to a $R$-derivation $\mathrm{d} \colon S \to S \oplus I[1]$. Conversely, a section d of $S \oplus I[1] \to S$ gives an extension by the cartesian square in $\mathrm{CAlg}^{\mathrm{an}}_R$:

$$\begin{array}{ccc} \tilde{S} & \longrightarrow & S \\ \downarrow & & \downarrow{\scriptstyle \mathrm{d}_0} \\ S & \xrightarrow[\mathrm{d}]{} & S \oplus I[1], \end{array}$$

where $\mathrm{d}_0 \colon S \to S \oplus I[1]$ is the trivial derivation.

*Remark* 2.91. In Theorem 2.74, the direction (b) implies (a) can be shown by the machinery we developed above. Consider the lifting problem in Remark 2.89. By the cartesian square in Remark 2.90, $\tilde{S} \to S$ admits an $R$-algebra section if and only if d is homotopic to $\mathrm{d}_0$. It suffices that

$$\pi_0 \operatorname{Map}_{\mathrm{D}_{\geq 0}(S)}(\mathrm{L}_{S/R}, I[1]) \simeq \operatorname{Ext}^1_S(\mathrm{L}_{S/R}, I) = 0.$$

This holds if $\mathrm{L}_{S/R}$ is a projective $S$-module, hence a retract of a free module $S^{(J)}$. Indeed,

$$\operatorname{Map}_S(S^{(J)}, I[1]) \simeq \prod_{j \in J} \operatorname{Map}_S(S, I[1]) \simeq \prod_{j \in J} I[1],$$

where the right side denotes a product of underlying anima and has trivial 0-truncation.

**Definition 2.92** (Regular immersions)**.** A closed immersion of schemes $Z \hookrightarrow X$ is *regular*,[22] if every $x \in X$ has an affine open neighborhood $x \in U \subseteq X$ admitting a Tor-independent cartesian square

$$\begin{array}{ccc} Z \cap U & \hookrightarrow & U \\ \downarrow & & \downarrow \\ \operatorname{Spec}(\mathbf{Z}) & \hookrightarrow & \mathbf{A}^n_{\mathbf{Z}}, \end{array}$$

where the horizontal maps are closed immersions. In terms of rings, this is a cocartesian square

$$\begin{array}{ccc} \mathbf{Z}[x_1, \ldots, x_n] & \longrightarrow & \mathcal{O}(U) \\ \downarrow & & \downarrow \\ \mathbf{Z} & \longrightarrow & \mathcal{O}(Z \cap U). \end{array}$$

Tor independence means that $\operatorname{Tor}_i^{\mathbf{Z}[x_1,\ldots,x_n]}(\mathbf{Z}, \mathcal{O}(U)) = 0$ for all $i > 0$. Equivalently, Tor-independence ensures that the above square is also cocartesian in animated rings.

*Example* 2.93 (Effective Cartier divisor)*.* For $n = 1$, Definition 2.92 gives the notion of an *effective Cartier divisor*. In this case, $Z \cap U \simeq \mathrm{V}(f)$ for some $f \in \mathcal{O}(U)$ and Tor independence of the square

$$\begin{array}{ccc} \mathrm{V}(f) & \hookrightarrow & U \\ \downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(\mathbf{Z}) & \hookrightarrow & \mathbf{A}^1_{\mathbf{Z}} \end{array}$$

is equivalent to $f$ being a nonzero divisor.

*Example* 2.94 (Regular closed immersion)*.*

(a) The composite of a chain of effective Cartier divisors

$$Z_n \hookrightarrow Z_{n-1} \hookrightarrow \cdots \hookrightarrow Z_1 \hookrightarrow X$$

[22] In the literature, this is also called *locally of complete intersection*, or *quasi-smooth*.

is a regular immersion.

(b) A noetherian local ring $(A, \mathfrak{m})$ is regular if and only if $\mathrm{Spec}(\kappa(\mathfrak{m})) \hookrightarrow \mathrm{Spec}(A)$ is regular.

**Notation 2.95.** Let $X$ be a scheme and $M \in \mathrm{QCoh}(X)$.

(a) the *affine space* of $M$ over $X$ is $\mathbf{V}(M) = \mathbf{A}(M) = \mathrm{Spec}_X(\mathrm{Sym}(M))$;

(b) the *projective space* of $M$ over $X$ is $\mathbf{P}(M) = \mathrm{Proj}_X(\mathrm{Sym}(M))$.

**Definition 2.96** (regular closed immersion)**.** Let $X$ be a scheme and let $M$ be a finite locally free sheaf on $X$. An $\mathcal{O}_X$-linear map $u\colon M \to \mathcal{O}_X$[23] is *regular* if the cartesian square

$$\begin{array}{ccc} \mathbf{V}(u) & \longrightarrow & X \\ \downarrow & & \downarrow \\ X & \xrightarrow[u]{} & \mathbf{V}(M) \end{array}$$

defining its zero locus $\mathbf{V}(u)$ is Tor independent, where the right vertical map is the zero section.

*Remark* 2.97.

(a) A closed immersion of schemes $Z \hookrightarrow X$ is regular in the sense of Definition 2.92 if and only if it is locally of the form $\mathbf{V}(u) \hookrightarrow U$ for a regular map $u\colon M \to \mathcal{O}_U$, where $M$ is a finite locally free sheaf on $U$.

(b) A closed immersion of schemes $Z \hookrightarrow X$ of finite presentation is regular if and only if $\mathrm{L}_{Z/X}[-1]$ is locally free. Equivalently, the conormal sheaf $\mathcal{N}_{Z/X}$ is locally free and $\mathrm{H}_i(\mathrm{L}_{Z/X}) = 0$ for every $i \geq 2$.

(c) An immersion of schemes $i\colon Z \hookrightarrow X$ is a *regular immersion*, if it admits a factorization $Z \overset{j}{\hookrightarrow} U \overset{k}{\hookrightarrow} X$, where $j$ is a closed regular immersion and $k$ is an open immersion.

(d) Let $Z \hookrightarrow X$ be a regular closed immersion of schemes with ideal sheaf $I$. Its *blowup* is

$$\mathrm{Bl}_Z(X) = \mathrm{Proj}_X\left(\bigoplus_{n\geq 0} I^n\right).$$

Both squares in the diagram

$$\begin{array}{ccccc} \mathbf{P}(C_{Z/X}) & \hookrightarrow & \mathbf{P}(\mathcal{N}_{Z/X}) & \longrightarrow & Z \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Bl}_Z(X) & \hookrightarrow & \mathbf{P}(I) & \longrightarrow & X \end{array}$$

are cartesian, where

$$\mathbf{P}(C_{Z/X}) = \mathrm{Proj}_Z\left(\bigoplus_{n\geq 0} I^n/I^{n+1}\right).$$

By regularity, the closed immersions $\mathrm{Bl}_Z(X) \hookrightarrow \mathbf{P}(I)$ and $\mathbf{P}(C_{Z/X}) \hookrightarrow \mathbf{P}(\mathcal{N}_{Z/X})$ are isomorphisms.

**Proposition 2.98.** *Let $S$ be a scheme. Every closed immersion in $\mathrm{Sm}_S$ is regular.*

*Proof.* Let $i\colon Z \hookrightarrow X$ be a closed immersion in $\mathrm{Sm}_S$, as in the diagram

$$\begin{array}{ccc} Z & \overset{i}{\hookrightarrow} & X \\ & \searrow & \downarrow \\ & & S. \end{array}$$

Since immersions are unramified, the fiber sequence of cotangent complexes

$$i^*\mathrm{L}_{X/S} \longrightarrow \mathrm{L}_{Z/S} \longrightarrow \mathrm{L}_{Z/X}$$

[23] By definition, $\mathcal{O}_X$-linear maps $M \to \mathcal{O}_X$ correspond to the sections of $\mathbf{V}(M) \to X$.

gives the short exact sequence

$$0 \longrightarrow \mathcal{N}_{Z/X} \longrightarrow i^*\Omega_{X/S} \longrightarrow \Omega_{Z/S} \longrightarrow 0$$

and $\mathrm{H}_i(\mathrm{L}_{Z/X}) = 0$ for every $i \geq 2$. Since $i^*\Omega_{X/S}$ and $\Omega_{Z/S}$ are locally free, so is $\mathcal{N}_{Z/X}$. By Remark 2.97, $i\colon Z \hookrightarrow X$ is a regular closed immersion. □

*Exercise* 2.99. Let $f\colon X \to S$ be a morphism of schemes locally of finite presentation. Show that $f$ is smooth if and only if it is flat and its diagonal $\Delta_f\colon X \to X \times_S X$ is a regular immersion.

**Proposition 2.100.** *Let $Z \hookrightarrow X$ be a closed immersion in $\mathrm{Sm}_S$ with ideal sheaf $I$. Suppose that $f_1, \ldots, f_n \in \mathcal{O}(X)$ cut out $Z$ and induce a basis of $\mathcal{N}_{Z/X} = I/I^2$. Then the map $(f_1, \ldots, f_n)\colon \mathcal{O}_X^n \to \mathcal{O}_X$ is regular.*

*Proof.* Since $Z \hookrightarrow X$ is a closed immersion in $\mathrm{Sm}_S$, there is the short exact sequence

$$0 \longrightarrow \mathcal{N}_{Z/X} \xrightarrow{\mathrm{d}} \Omega_{X/S}|_Z \longrightarrow \Omega_{Z/S} \longrightarrow 0.$$

Locally,[24] we can extend $f_1, \ldots, f_n$ to functions $f_1, \ldots, f_m$ whose differentials form a basis of $\Omega_{X/S}$, and the map $f = (f_1, \ldots, f_m)\colon X \to \mathbf{A}_S^m$ induces an isomorphism $f^*\mathrm{L}_{\mathbf{A}_S^m/S} \xrightarrow{\sim} \mathrm{L}_{X/S}$, whose cofiber $\mathrm{L}_{X/\mathbf{A}_S^m}$ vanishes, so $f$ is étale. The commutative squares

$$\begin{array}{ccccc} Z & \longrightarrow & \mathbf{A}_S^{m-n} & \longrightarrow & S \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle 0} \\ X & \xrightarrow[f]{} & \mathbf{A}_S^m & \xrightarrow[\pi]{} & \mathbf{A}_S^n \end{array}$$

are cartesian, where $\pi$ is projection onto the first $n$ coordinates. Since $f$ and $\pi$ are flat, the outer rectangle is Tor independent. Consider the cartesian squares

$$\begin{array}{ccccc} Z & \longrightarrow & X & \longrightarrow & S \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle 0} \\ X & \longrightarrow & \mathbf{V}(\mathcal{O}_X^n) & \longrightarrow & \mathbf{A}_S^n. \end{array}$$

The morphism $X \to \mathbf{V}(\mathcal{O}_X^n)$ is the base change of $X \to S$ along $\mathbf{A}_S^n \to S$, hence is flat. Therefore, by the pasting law for Tor-independent squares, the left square is Tor-independent. Since Tor-independence is local on the base, the map$(f_1, \ldots, f_n)\colon \mathcal{O}_X^n \to \mathcal{O}_X$ is regular. □

**Lemma 2.101.** *Let $A \to B$ be a local map of local rings, essentially of finite presentation, $\mathfrak{m}$ the maximal ideal of $A$ and let $M$ be a finitely presented $B$-module. If $\mathrm{Tor}_1^A(A/\mathfrak{m}, M) = 0$, then $M$ is flat over $A$.*

*Proof sketch.* Reduce to the noetherian case and apply the Artin–Rees lemma; see [Sta26, Tag 0471]. □

**Theorem 2.102** (Fiberwise regularity implies flatness)**.** *Let $A \to B$ be a local map of local rings, essentially of finite presentation, $P$ a finitely generated projective $B$-module, and let $u\colon P \to B$ be a $B$-linear map. If $B$ is flat over $A$ and*

$$\bar{u}\colon P/\mathfrak{m}P \longrightarrow B/\mathfrak{m}B$$

*is regular, where $\mathfrak{m}$ is the maximal ideal of $A$, then $B/u(P)$ is flat over $A$.*

*Proof.* See [GD67][Part 3, Theorem 11.3.8.]. Since $B$ is local, choose $P \simeq B^n$ and write $u = (u_1, \ldots, u_n)$ and $I = u(P)$. Let $R = \mathbf{Z}[x_1, \ldots, x_n]$ and choose a projective resolution $K_\bullet \to \mathbf{Z}$ over $R$ with $K_0 = R$, $K_1 = R^n$, and differential $(x_1, \ldots, x_n)\colon K_1 \to K_0$; the augmentation $R \to \mathbf{Z}$ sends each $x_i$ to 0. The $R$-algebra structure of $B$ is given by $R \to B$, $x_i \mapsto u_i$. Applying $- \otimes_R B$ to $K_\bullet$ gives us the augmented complex

$$K_2 \otimes_R B \xrightarrow{\mathrm{d}} P \xrightarrow{u} B \twoheadrightarrow B/I \longrightarrow 0,$$

[24] In the following, we will abuse the notations and work locally. Moreover, the morphism $f$, the natural number $m \in \mathbf{N}$ and the isomorphism of cotangent complexes are all intermediate constructions.

hence $B/I$ is finitely presented over $B$. Moreover, applying $- \otimes_B A/\mathfrak{m}$ gives us the complex

$$K_2 \otimes_R (B/\mathfrak{m}B) \xrightarrow{\bar{\mathrm{d}}} P/\mathfrak{m}P \xrightarrow{\bar{u}} B/\mathfrak{m}B.$$

Furthermore, by right exactness of $- \otimes_B A/\mathfrak{m}$, there are surjections

$$P/\mathfrak{m}P \twoheadrightarrow \operatorname{coker}(\bar{\mathrm{d}}) \twoheadrightarrow I/\mathfrak{m}I \twoheadrightarrow \operatorname{im}(\bar{u}).$$

By the pasting law for Tor-independent squares, regularity of $\bar{u}$ implies that $\operatorname{Tor}_i^R(\mathbf{Z}, B/\mathfrak{m}B) = 0$ for $i > 0$. Hence $K_\bullet \otimes_R (B/\mathfrak{m}B)$ is exact in positive degrees, and the composite $\operatorname{coker}(\bar{\mathrm{d}}) \to \operatorname{im}(\bar{u})$ is an isomorphism. It follows that $I/\mathfrak{m}I \to B/\mathfrak{m}B$ is injective. The short exact sequence $I \hookrightarrow B \twoheadrightarrow B/I$ induces the exact sequence

$$\operatorname{Tor}_1^A(A/\mathfrak{m}, B) \longrightarrow \operatorname{Tor}_1^A(A/\mathfrak{m}, B/I) \longrightarrow I/\mathfrak{m}I \longrightarrow B/\mathfrak{m}B.$$

By flatness of $B$ over $A$, $\operatorname{Tor}_1^A(A/\mathfrak{m}, B) = 0$, hence $\operatorname{Tor}_1^A(A/\mathfrak{m}, B/I) = 0$. Since $B/I$ is finitely presented over $B$, Lemma 2.101 implies that $B/u(P)$ is flat over $A$. □

**Definition 2.103** (Étale neighborhoods)**.** Let $Z \hookrightarrow X$ be a closed immersion of schemes. An *étale neighborhood* of $Z$ in $X$ is an étale morphism $V \to X$ such that the projection $V \times_X Z \to Z$ is an isomorphism.

**Theorem 2.104** (Lifting of vector bundles)**.** *Let $Z \hookrightarrow X$ be a closed immersion of affine schemes and let $M$ be a finite locally free sheaf on $Z$. There exist an affine étale neighborhood $V \to X$ of $Z$ and a finite locally free sheaf $\widehat{M}$ on $V$ such that $\widehat{M}|_Z \simeq M$.*

*Proof.* We will give an elementary proof of this statement. See [Sta26, Tag 07M5] for an alternative proof. Write $X = \operatorname{Spec}(A)$ and $Z = \operatorname{Spec}(A/I)$, and let $\bar{P}$ be the finitely generated projective $A/I$-module corresponding to $M$. Indeed, $P \simeq \operatorname{im}(\bar{e})$, where $\bar{e} \in \operatorname{Mat}_{n\times n}(A/I)$ is an idempotent. Choose an arbitrary lift $e \in \operatorname{Mat}_{n\times n}(A)$, which is not necessarily an idempotent.

Write $I = \operatorname{colim}_{j\in J} I_j$ as the filtered union of its finitely generated subideals $I_j$. Since the entries of $e^2 - e$ lie in $I$, it is zero in $\operatorname{Mat}_{n\times n}(A/I_j)$ for some $j \in J$. We may therefore assume that $I = (\lambda_1, \dots, \lambda_m)$ is finitely generated. Every lift of $\bar{e}$ takes the form $e + \sum_{i=1}^m \lambda_i \alpha_i = \lambda\alpha$,[25] where $\alpha_i \in \operatorname{Mat}_{n\times n}(A)$. If $e^2 - e$ is nonzero, we lift the entries of $e^2 - e$ along $\lambda\colon A^m \to I$ as in the diagram

$$\begin{array}{ccc} & & A^m \\ & \overset{\beta}{\nearrow} & \big\downarrow \\ A^{n^2} & \xrightarrow[e^2-e]{} & I. \end{array}$$

Writing the components of $\beta$ as matrices $\beta_1, \dots, \beta_m \in \operatorname{Mat}_{n\times n}(A)$ gives $e^2 - e = \sum_{i=1}^m \lambda_i \beta_i$.

We will now geometrize this lifting problem. Indeed, $e$ and $\beta$ are fixed, and $\alpha$ is a variable to solve. Since $\alpha$ is a $m$-tuple of $(n \times n)$-matrices, $\mathbf{A}_A^{mn^2}$ is the suitable parametrization. By the computation

$$(e + \lambda\alpha)^2 - (e + \lambda\alpha) + [e, \lambda\alpha] = \lambda(\beta + (2e - \mathrm{id})\alpha + \alpha(\lambda\alpha)),$$

we find that $e$ is an idempotent if the right hand side vanishes and $[e, \lambda\alpha] = 0$, where $[e, \lambda\alpha] = e(\lambda\alpha) - (\lambda\alpha)e$.

Consider the closed subscheme $V = \mathbf{V}(\beta + (2e - \mathrm{id})\alpha) \subseteq \mathbf{A}_A^{nm^2}$. We claim that $V$ is a neighborhood of $Z$, i.e., $\bar{V} = V \times_{\operatorname{Spec}(A)} \operatorname{Spec}(A/I) \simeq \operatorname{Spec}(A/I)$. Indeed, by setting $\lambda = 0$, we see that $\bar{V} = \mathbf{V}(\beta + (2e - \mathrm{id})\alpha)$. Since $e$ is idempotent, $(2e - \mathrm{id})$ is invertible and $\alpha$ has a unique solution. By elimination of variables,[26] $\bar{V} \simeq \operatorname{Spec}(A/I)$.

We first show that there exists $f \in A$, whose image in $A/I$ is a unit, such that $[e, \lambda\alpha] = 0$, after localizing at $f$, we obtain a neighborhood of $Z$ on which $e + \lambda\alpha$ is an idempotent. We again denote this by $V$. Indeed, take $f = \det(2e - \mathrm{id}) \in A$. In $A/I$, the matrix $(2e - \mathrm{id})$ is invertible, hence $f$ is a unit. After localizing at $f$, we may

[25] Here, $\alpha$ and $\lambda$ collect the $\alpha_i$ and $\lambda_i$, respectively, so that $\lambda\alpha$ is simply matrix multiplication. For simplicity, we adopt this convention and slightly abuse the notation in the following to write $\alpha(\lambda\alpha) = (\sum_{j=1}^m \lambda\alpha_1\alpha_j, \dots, \sum_{j=1}^m \lambda\alpha_m\alpha_j)$.

[26] This is the matrix analogue of $R[x]/(ux + c) \simeq R$, where $u \in R^\times$ and $c \in R$.

assume that $(2e - \mathrm{id})$ is invertible. For the commutator, we compute

$$0 = [e, \lambda\beta + (2e - \mathrm{id})\lambda\alpha + (\lambda\alpha)^2] = (2e - \mathrm{id})[e, \lambda\alpha] + [e, (\lambda\alpha)^2],$$

which follows from $[e, \lambda\beta] = [e, e^2 - e] = 0$. This implies that

$$[e, \lambda\alpha] = -(2e - \mathrm{id})^{-1}[e, (\lambda\alpha)^2] = -\lambda(2e - \mathrm{id})^{-1}(\alpha[e, \lambda\alpha] + [e, \lambda\alpha]\alpha) = \lambda M([e, \lambda\alpha]),$$

where $M \in \mathrm{End}(\mathcal{O}(\mathbf{A}^{nm^2}_A)^n)$. In other words, $(\mathrm{id} - \lambda M)([e, \lambda\alpha]) = 0$. Therefore, we may take $f = \det(\mathrm{id} - \lambda M)$. Obviously, $f \equiv 1 \mod I$, which is invertible. In particular, $\mathrm{id} - \lambda M$ is also invertible, hence $[e, \lambda\alpha] = 0$.

Finally, we show that there exists $f \in A$, whose image in $A/I$ is a unit, such that after shrinking $V$ further by localizing at $f$, $V$ becomes étale over $A$. Indeed, take $\mathrm{J} = \mathrm{Jac}(\beta + (2e - \mathrm{id})\alpha + \alpha(\lambda\alpha))$. Since $\lambda\alpha^2 \equiv 0 \mod I$, $\beta$ is constant and $(2e - \mathrm{id})$ is invertible, J is the diagonal matrix $\mathrm{diag}((2e - \mathrm{id}), \ldots, (2e - \mathrm{id}))$ with $nm$-blocks. Taking $f = \det(J) \in A$ and passing to the localization at $f$, we see that the Jacobian criterion for étaleness is satisfies, hence $V$ is étale over $A$. □

**Theorem 2.105** (Lifting of smooth closed subschemes)**.** *Consider a cartesian square of affine schemes*

$$\begin{array}{ccc} X_0 & \hookrightarrow & X \\ \downarrow & & \downarrow \\ S_0 & \hookrightarrow & S, \end{array}$$

*where $S_0 \hookrightarrow S$ is a closed immersion and $X \to S$ is smooth. Let $Z_0 \hookrightarrow X_0$ be a closed immersion with $Z_0 \to S_0$ smooth. There exist an affine étale neighborhood $V \to X$ of $Z_0$ and a closed subscheme $Z \hookrightarrow V$, smooth over $S$, such that $Z \times_S S_0 \simeq Z_0$.*

*Proof.* The idea is to lift the equations cutting out $Z_0$ to $X$. Their zero locus will be smooth near $Z_0$.

Since $X_0$ and $Z_0$ are smooth over $S_0$, by Proposition 2.98, the closed immersion $Z_0 \hookrightarrow X_0$ is regular. Its conormal sheaf $\mathcal{N}_{Z_0/X_0}$ is therefore finite locally free. By Theorem 2.104, there exist an affine étale neighborhood $V \to X$ of $Z_0$ and a finite locally free sheaf $M$ on $V$ with $M|_{Z_0} \simeq \mathcal{N}_{Z_0/X_0}$. Put $V_0 = V \times_X X_0$, $M_0 = M|_{V_0}$.

$$\begin{array}{ccc} Z_0 & \hookrightarrow & V_0 \\ {\scriptstyle \mathrm{id}_{Z_0}}\downarrow & & \downarrow \\ Z_0 & \hookrightarrow & X_0 \end{array}$$

is cartesian. Let $I$ be the ideal sheaf of $Z_0$ in $V_0$ and the fundamental exact sequence identifies

$$\mathcal{N}_{Z_0/X_0} \simeq \mathcal{N}_{Z_0/V_0} \simeq I/I^2.$$

We have surjections $M \twoheadrightarrow M_0$ and $I \twoheadrightarrow I/I^2$, viewing sheaves on closed subschemes as sheaves on the ambient scheme by pushforward. Since $V$ and $V_0$ are affine, $M$ is projective in $\mathrm{QCoh}(V)$ and $M_0$ is projective in $\mathrm{QCoh}(V_0)$. We can therefore lift $M_0 \to I/I^2$ to $\varphi_0 \colon M_0 \to I$, and obtain $\varphi \colon M \to \mathcal{O}_V$:

$$\begin{array}{ccc} & & I \\ & {\scriptstyle \exists \varphi_0} \nearrow & \downarrow \\ M_0 & \longrightarrow & \mathcal{N}_{Z_0/X_0}, \end{array} \qquad \begin{array}{ccccc} & & & & \mathcal{O}_V \\ & & {\scriptstyle \varphi} \nearrow & & \downarrow \\ M & \twoheadrightarrow & M_0 & \xrightarrow{\varphi_0} & I \hookrightarrow \mathcal{O}_{V_0}. \end{array}$$

Let $Z = \mathbf{V}(\varphi) \hookrightarrow V$. After shrinking $V$, we must show that $Z \times_S S_0 \simeq Z_0$ and that $Z \to S$ is smooth. First,

$$Z \times_S S_0 = Z \times_V V_0 = \mathbf{V}(\varphi_0).$$

Since $M|_{Z_0} \simeq \mathcal{N}_{Z_0/X_0}$, the restriction of $\varphi_0 \colon M_0 \to I$ induces the an isomorphism

$$M_0 \otimes_{\mathcal{O}_{V_0}} \mathcal{O}_{Z_0} \xrightarrow{\sim} I/I^2.$$

Since $I$ is finitely generated, Nakayama's lemma implies that $\varphi_0$ is surjective in an open neighborhood of $Z_0$. Therefore, $Z_0 \subseteq \mathbf{V}(\varphi_0)$ is clopen and $\mathbf{V}(\varphi_0) = Z_0 \sqcup W$, where $W = \mathbf{V}(\varphi_0) \setminus Z_0$. Under the surjection

$$\mathcal{O}(V) \twoheadrightarrow \mathcal{O}(Z_0) \times \mathcal{O}(W),$$

choose $f \in \mathcal{O}(V)$ mapping to $(1, 0)$. Replacing $V$ by $V_f$ removes $W$ and gives $Z \times_S S_0 \simeq Z_0$.

We will shrink $V$ further for $Z \to S$ to be smooth. Consider the cartesian square

$$\begin{array}{ccc} Z_0 & \hookrightarrow & Z \\ \downarrow & & \downarrow \\ S_0 & \hookrightarrow & S. \end{array}$$

Since $Z_0 \to S_0$ is smooth, the fibers of $Z \to S$ over $S_0$ are smooth. The morphism $Z \to S$ is locally of finite presentation, hence the fiberwise criterion reduces smoothness at $z \in Z_0$ to flatness at this point.

Fix $z \in Z_0$ with image $s \in S_0$. We must show that $\mathcal{O}_{Z,z}$ is flat over $\mathcal{O}_{S,s}$. The ring $\mathcal{O}_{V,z}$ is flat over $\mathcal{O}_{S,s}$ because $V \to S$ is smooth. By Theorem 2.102, it suffices to show that $\bar{\varphi}\colon M|_{V_s} \to \mathcal{O}_{V_s}$ is regular, where $V_s = V \times_S \operatorname{Spec}(\kappa(s))$. The map $\varphi_0\colon M_0 \to \mathcal{O}_{V_0}$ cuts out $Z_0$ and $\operatorname{rk}(M_0) = \operatorname{rk}(\mathcal{N}_{Z_0/V_0})$, hence by Proposition 2.100, $\varphi_0$ is regular. In particular, $\bar{\varphi} = \varphi_0 \otimes_{S_0} \kappa(s)$ is regular. This proves the required flatness, hence smoothness at $z \in Z_0$. Since the smooth locus of a morphism locally of finite presentation is open, shrinking $V$ and restricting $Z$ gives the smooth morphism $Z \to S$. By Exercise 2.109, we may shrink $V$ futher so that it is affine.

□

**Corollary 2.106.** *Let $X_0 \to S_0$ be a smooth morphism of affine schemes and let $S_0 \hookrightarrow S$ be a closed immersion of affine schemes. There exists a smooth affine $S$-scheme $X$ such that $X \times_S S_0 \simeq X_0$.*

*Proof.* Choose a closed immersion $X_0 \hookrightarrow \mathbf{A}^n_{S_0}$ and apply Theorem 2.105 to the cartesian square

$$\begin{array}{ccc} \mathbf{A}^n_{S_0} & \hookrightarrow & \mathbf{A}^n_S \\ \downarrow & & \downarrow \\ S_0 & \hookrightarrow & S. \end{array}$$

This produces the desired smooth affine $S$-scheme $X$. □

**Corollary 2.107** (Smooth closed subschemes are neighborhood retracts)**.** *Consider a commutative triangle of affine schemes*

$$\begin{array}{ccc} Z & \hookrightarrow & X \\ & \searrow & \downarrow \\ & & S, \end{array}$$

*where $Z \hookrightarrow X$ is a closed immersion and $Z \to S$ is smooth. There exists an affine étale neighborhood $V \to X$ of $Z$ such that $Z \hookrightarrow V$ admits a retraction over $S$.*

*Proof.* Consider the cartesian square

$$\begin{array}{ccc} Z \times_S Z & \hookrightarrow & X \times_S Z \\ \downarrow & & \downarrow \\ Z & \hookrightarrow & X, \end{array}$$

where the vertical maps are smooth and the horizontal maps are closed immersions. Apply Theorem 2.105 to the diagonal $\Delta_Z\colon Z \hookrightarrow Z \times_S Z$. We obtain an affine étale neighborhood $W \to X \times_S Z$ of $\Delta_Z$ and a closed subscheme $V \hookrightarrow W$, smooth over $X$, such that $V \times_X Z \simeq \Delta_Z$.

The rank of $\Omega_{V/X}$ is locally constant and vanishes along $Z$. Replacing $V$ by the clopen subscheme where this rank is 0, we may therefore assume that $V \to X$ is étale. It is then an affine étale neighborhood of $Z$ in $X$. In

$$\begin{array}{ccccc} Z & \xrightarrow{\Delta_Z} & Z \times_S Z & \xrightarrow{\pi_2} & Z, \\ & \searrow^{f} & \downarrow & \nearrow_{\pi_2} & \\ & & X \times_S Z & & \end{array}$$

the morphism $f$ is the composite $Z \hookrightarrow V \hookrightarrow W \to X \times_S Z$. Since $\pi_2 \circ \Delta_Z = \mathrm{id}$, the composite $V \to W \to X \times_S Z \to Z$ is the desired retraction. □

**Theorem 2.108** (Tubular neighborhood theorem)**.** *Let $Z \hookrightarrow X$ be a closed immersion in $\mathrm{Sm}_S$, with $Z$, $X$, and $S$ affine. There is a zigzag*

$$X \leftarrow V \to \mathbf{V}(\mathcal{N}_{Z/X}),$$

*where $V$ is an affine étale neighborhood both of $Z$ in $X$ and of the zero section $Z \hookrightarrow \mathbf{V}(\mathcal{N}_{Z/X})$ of the normal bundle.*

*Proof sketch.* todo

□

*Exercise* 2.109. In the setting of Theorem 2.108, show the following.

(a) Let $h\colon Y \to X$ be étale, with compatible closed immersions $Z \hookrightarrow Y$ and $Z \hookrightarrow X$. There is an open neighborhood $U \subseteq Y$ of $Z$ such that $h|_U$ is an étale neighborhood of $Z$ in $X$.

(b) Let $Z \hookrightarrow X$ be a closed immersion of affine schemes and let $U \subseteq X$ be an open neighborhood of $Z$. There is an affine open neighborhood $V \subseteq U$ of $Z$.

## 3. Localization and purity

## 4. Classifying spaces

# 5. Vector bundles

## 6. Motivic spectra

## 7. Examples

## 8. Orientations and cobordism

## 9. The six-functor formalism

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